

QUANTUM TURING PATTERNS

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ABSTRACT. We construct quantum Turing patterns in Lindblad lattice dynamics and establish a rigorous theory of their nonlinear order and quantum fluctuations. For an explicit family of completely positive lattice generators with finite-range couplings, the first-moment equations undergo a supercritical instability at a nonzero wave number and admit analytic site- and bond-centered commensurate stripe branches. These branches are locally asymptotically stable in their reflection-fixed period-cell spaces, and projected coherent states exhibit extensive Bragg order on every bounded time interval in the semiclassical limit. We prove $O(N^{-1/2})$ convergence of microscopic covariances to a nonautonomous Gaussian Lyapunov flow, transferring strict partial-transpose uncertainty violations to sufficiently large N . In the homogeneous Gaussian sector, a single dimensionless ratio controls both the Turing stability determinant and the logarithmic negativity of opposite momenta, relating wavelength selection directly to quantum entanglement. Differential transport shifts the strongest opposite-momentum correlations from the infrared to the selected Turing scale. Numerical continuation and two-dimensional simulations display stripe, spot, and labyrinth morphologies whose Fourier modes and fluctuation spectra concentrate at the same selected wave numbers.

1. INTRODUCTION AND MAIN RESULTS

Classical Turing patterns arise when differential transport destabilizes a homogeneous state at a nonzero wave number while the homogeneous mode remains stable. The mechanism organizes stripes, spots, and labyrinths across reaction–diffusion theory and experiment [1–6]. Its biological role is supported by pigment rearrangement, hair-follicle spacing, digit specification, palatal-ruga formation, and systematic searches of biochemical networks [5, 7–14]. These developments sharpen a more microscopic question: can the same pattern-selection mechanism emerge from quantum Markov dynamics?

A *quantum Turing pattern* is a Turing pattern generated by Lindblad lattice dynamics and detected by microscopic noncommuting observables. We then determine how the selected wavelength appears in its quantum fluctuation spectrum.

Quantum systems have displayed Turing-type symmetry breaking in several settings. Levitov et al. predicted a triangular modulation in a spatially uniform two-dimensional exciton system, while Ardizzone et al. observed and controlled transverse Turing patterns in a coherent polariton fluid [15, 16]. In open quantum systems, Bandyopadhyay et al. considered two coupled quantum oscillators, Kato and Nakao studied two quantum activator–inhibitor units, and Comparato et al. analyzed mode competition in a finite bosonic GKSL chain [17–19]. Chia et al. addressed Lindblad quantization of planar polynomial flows [20]. These studies consider quantum-fluid patterns, few-mode open-system instabilities, and general Lindblad quantization, but none derives the critical wave number, nonlinear spatial branch, and microscopic fluctuation dynamics within a single lattice construction.

To our knowledge, this is the first rigorous constructive theory of quantum Turing patterns in a Lindblad lattice. Using only finite-range couplings, the construction fixes a nonzero critical wave number and proves the associated nonlinear branch and volume-uniform Bragg order. It also establishes covariance convergence for the microscopic dynamics and derives a closed relation between the Turing stability determinant and opposite-momentum entanglement. The explicit family yields a lattice-commensurate stripe; an isotropic specialization produces two-dimensional spot and labyrinth morphologies whose fluctuation spectra concentrate on the selected momentum shell.

Date: Aug 3, 2026.

2020 Mathematics Subject Classification. Primary 81S22; Secondary 35B36, 37G40, 81P40, 82C10.

Theorem 1.1 (Existence, stability, and Bragg order for quantum Turing patterns). *There exists an explicit family of translation-invariant Lindblad generators with finite-range couplings on two-dimensional bosonic lattices with the following properties.*

- (i) *The first-moment limit undergoes a supercritical Turing bifurcation at a nonzero lattice wave number and admits analytic site- and bond-centered commensurate stripe branches.*
- (ii) *The two branches are locally asymptotically stable in their respective reflection-fixed period-cell spaces.*
- (iii) *For projected coherent product states prepared on either branch, the quantum structure factor has an extensive Bragg peak on every bounded time interval in the semiclassical limit.*

We denote the selected stripe wave number by k_* . Section 3 gives the explicit coefficients, selected wave number, and branch asymptotics.

Beyond the explicit family, we prove two general results. Fix a lattice torus, a finite horizon $T < \infty$, and a bounded first-moment trajectory. Let $V_N(t)$ denote the microscopic covariance and $V(t)$ its Gaussian Lyapunov limit. Then

$$\sup_{0 \leq t \leq T} \|V_N(t) - V(t)\| = O(N^{-1/2}).$$

For a homogeneous Gaussian momentum pair, let κ_k denote its effective damping, g_k its squeezing amplitude, and Ω_λ its rotation frequency, and write

$$R_k := \sqrt{\kappa_k^2 + 4\Omega_\lambda^2}, \quad \eta_k := \frac{2|g_k|}{R_k}.$$

The pair is stable exactly when $\eta_k < 1$; in that regime,

$$\tilde{\nu}_-(k, -k) = \frac{1}{2(1 + \eta_k)}, \quad E_{\text{LN}}(k, -k) = \log_2(1 + \eta_k).$$

Here $\tilde{\nu}_-$ is the smaller partially transposed symplectic eigenvalue and E_{LN} is the logarithmic negativity.

The accompanying Lean 4 formalization verifies the algebraic identities entering the spectral design, Lindblad parameter map, and homogeneous formulas; the source is included in the author's GitHub repository <https://github.com/IKEDAKAZUKI/Quantum-Turing-Pattern>.

2. QUANTUM MARKOV DYNAMICS AND SEMICLASSICAL ESTIMATES

2.1. Truncated dynamics and coherent states. Let $d, L, N \in \mathbb{N}$ with $d, L, N \geq 1$, let e_j be the j th coordinate vector of $\mathbb{Z}^d$, and set $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$. The nearest-neighbor graph distance on this periodic lattice is denoted by d_L , and

$$B_r(x) := \{y \in \Lambda_L : d_L(x, y) \leq r\}$$

is its closed graph ball. Let a be the canonical annihilation operator on $\ell^2(\mathbb{N})$, let $n = a^*a$, and let

$$\mathbf{P}_{\leq M} := \mathbf{1}_{\{n \leq M\}}$$

be the spectral projection of n onto occupations $0, \dots, M$. More generally, for a self-adjoint operator A and a Borel set E , $\mathbf{1}_E(A)$ denotes its spectral projection; expressions such as $\mathbf{1}_{\{n_x \geq m\}}$ use this functional calculus.¹ For the Fock cutoff $M_N = \lfloor cN \rfloor$, $c > 0$, set

$$\begin{aligned} \mathcal{H}_N &= \mathbf{P}_{\leq M_N} \ell^2(\mathbb{N}), & a_{M_N} &= \mathbf{P}_{\leq M_N} a \mathbf{P}_{\leq M_N}, \\ \mathcal{H}_{L,N} &= \bigotimes_{x \in \Lambda_L} \mathcal{H}_N, & \mathfrak{A}_{L,N} &= B(\mathcal{H}_{L,N}). \end{aligned}$$

¹Throughout, A^* denotes the Hilbert-space adjoint; hence a_x^* is the creation operator and corresponds to $a_x^\dagger$ in common physics notation. For superoperators, $*$ denotes the trace-dual (Heisenberg) adjoint.

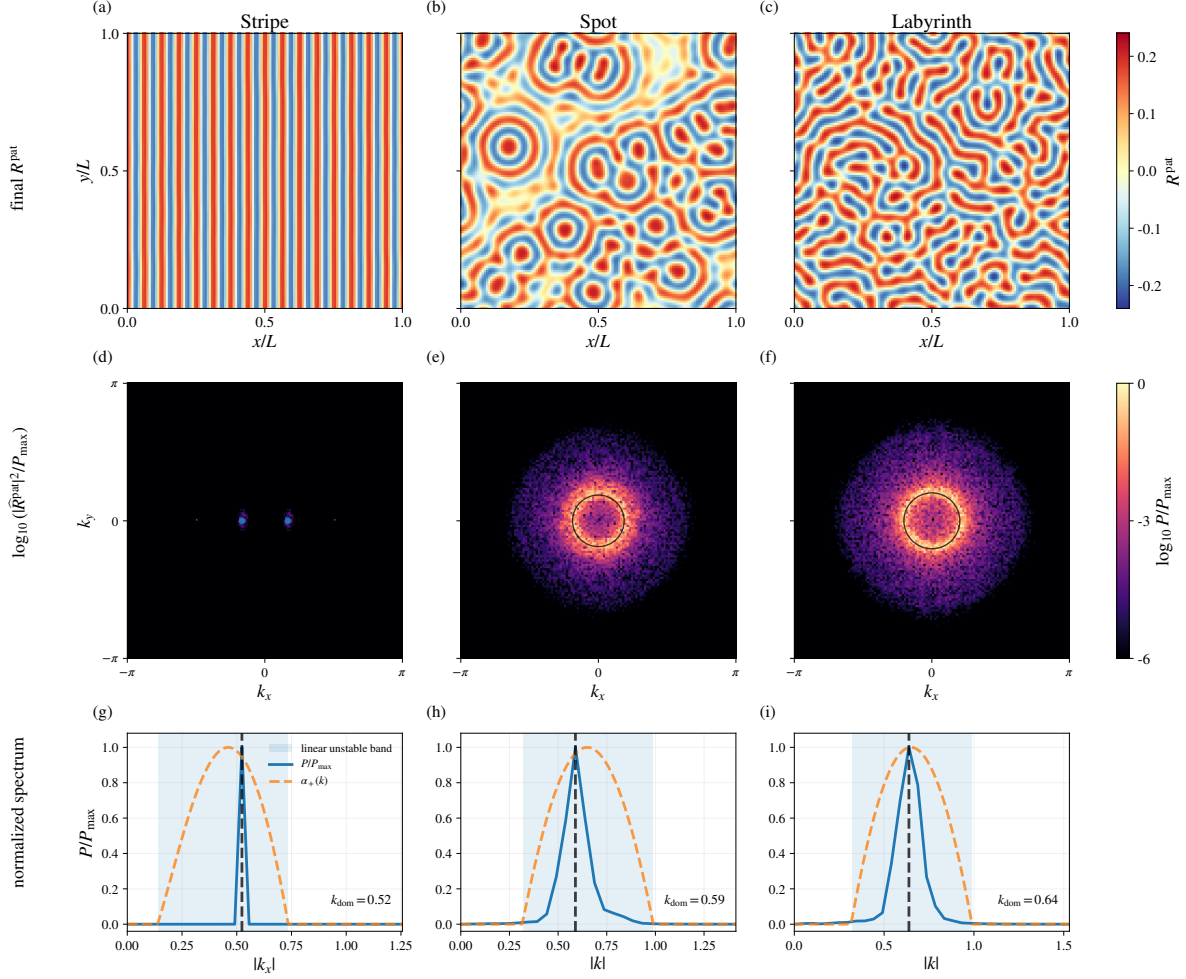


FIGURE 1. Quantum Turing morphologies and selected wave numbers in one Lindblad family. The top row shows the first-moment fields, the middle row their two-dimensional Fourier power, and the bottom row the selected spectra together with the linear unstable bands. The stripe selects the pair $\pm k_*$, while the isotropic spot and labyrinth regimes concentrate power on the selected wave-number shell.

The copies at $x \in \Lambda_L$ are denoted by a_{x,M_N} and $n_{x,M_N} = a_{x,M_N}^* a_{x,M_N}$; when the cutoff is fixed we abbreviate $n_x := n_{x,M_N}$. For a density matrix ρ and an observable A , write

$$\rho(A) := \text{Tr}(\rho A), \quad \rho_{\mathbf{m},\mathbf{n}} := \langle \mathbf{m} | \rho | \mathbf{n} \rangle, \quad [s]_+ := \max\{s, 0\},$$

where $\mathbf{m}, \mathbf{n} \in \{0, \dots, M_N\}^{\Lambda_L}$ are cutoff occupation multi-indices. We use standard notation for quantum Markov semigroups and Lindblad generators [21–23]. For density matrices we use the convention

$$\mathcal{D}[L]\rho = L\rho L^* - \frac{1}{2}\{L^*L, \rho\},$$

whose Heisenberg adjoint is $\mathcal{D}[L]^*(A) = L^*AL - \frac{1}{2}\{L^*L, A\}$. The Heisenberg generator has Lindblad form

$$(2.1) \quad \mathcal{L}_{L,N}(A) = i[H_{L,N}, A] + \sum_{\kappa} \left(V_{\kappa}^* A V_{\kappa} - \frac{1}{2} \{V_{\kappa}^* V_{\kappa}, A\} \right).$$

For an initial state $\rho_{L,N}(0)$, its Schrödinger evolution and the corresponding expectation shorthand are

$$\rho_{L,N}(t) := e^{t\mathcal{L}_{L,N}^*} \rho_{L,N}(0), \quad \rho_t(A) := \rho_{L,N}(t)(A), \quad \rho_0(A) := \rho_{L,N}(0)(A).$$

For every lattice direction, write

$$\Delta_j f_x = f_{x+e_j} + f_{x-e_j} - 2f_x, \quad j = 1, \dots, d.$$

Define the one-site rescaled quadratures and their copies by

$$Q_N = \frac{a_{M_N} + a_{M_N}^*}{\sqrt{2N}}, \quad P_N = \frac{a_{M_N} - a_{M_N}^*}{i\sqrt{2N}}, \quad Q_{x,N}, P_{x,N} \text{ at site } x.$$

Since

$$(2.2) \quad [a_{M_N}, a_{M_N}^*] = I - (M_N + 1)|M_N\rangle\langle M_N|,$$

operator norm is not the correct topology for the oscillator cutoff. We use a local coherent-state seminorm.

Definition 2.1. For $E < c$, let

$$\psi_\alpha^{(L,N)} = \bigotimes_{x \in \Lambda_L} \frac{\mathbf{P}_{\leq M_N} |\sqrt{N}\alpha_x\rangle}{\|\mathbf{P}_{\leq M_N} |\sqrt{N}\alpha_x\rangle\|}, \quad \sup_x |\alpha_x|^2 \leq E.$$

Let $\mathcal{S}_{L,N}(E)$ be the convex hull of the rank-one states $|\psi_\alpha^{(L,N)}\rangle\langle\psi_\alpha^{(L,N)}|$. For $A \in \mathfrak{A}_{L,N}$ set

$$\|A\|_{\mathcal{S}_{L,N}(E)} = \sup_{\rho \in \mathcal{S}_{L,N}(E)} \rho(A^*A)^{1/2}.$$

Lemma 2.2. For every $E < c$ there are constants $C, c_0 > 0$, independent of L , such that

$$\rho(|M_N\rangle_x \langle M_N|) \leq C e^{-c_0 N}, \quad \rho \in \mathcal{S}_{L,N}(E), \quad x \in \Lambda_L.$$

$$(2.3) \quad \|N[Q_{x,N}, P_{x,N}] - iI\|_{\mathcal{S}_{L,N}(E)} \leq C N e^{-c_0 N} \rightarrow 0.$$

Proof. The occupation number in $|\sqrt{N}\alpha\rangle$ is Poisson with mean at most NE . Since $M_N = \lfloor cN \rfloor$ and $c > E$, the Chernoff bound gives an exponentially small cutoff tail. Equation (2.3) follows from (2.2). The product-state estimate is uniform in L because the seminorm is defined through local coherent-state moments and the commutator is local. $\square$

Lemma 2.3. Fix a polynomial degree and a coherent energy bound $E < c$. For every local polynomial $\mathcal{P}$ in the rescaled variables $a_{x,M_N}/\sqrt{N}$ and $a_{x,M_N}^*/\sqrt{N}$, normal ordering changes its coherent symbol by $O_{\mathcal{S}_{L,N}(E)}(N^{-1})$, uniformly in L and in the site labels appearing in $\mathcal{P}$. The coherent first-moment equations obtained from (2.1) are classical polynomial symbol equations with $O_{\mathcal{S}_{L,N}(E)}(N^{-1})$ remainders.

Proof. On the coherent-state set of Definition 2.1, the cutoff tail is exponentially small by Lemma 2.2. Away from that tail the rescaled canonical commutator is N^{-1} times the identity up to an exponentially small error. Moving creation operators past annihilation operators in a fixed-degree polynomial produces only finitely many commutator terms, each smaller by one power of N . $\square$

2.2. Semiclassical estimates. Throughout the uniform semiclassical estimates below, $\mathcal{L}_{L,N}$ is a finite-range Lindblad generator with a fixed interaction range $r_0 \in \mathbb{N}$. Its coefficients are independent of L, N , and the cutoff boundary, and all scalar or matrix coefficients are uniformly bounded. Its terms are: (i) on-site quadratic Hamiltonians, with optional coherent drives $\sqrt{N}(f_x a_x^* + \bar{f}_x a_x)$ satisfying $\sup_{L,N,x} |f_x| < \infty$; (ii) directional quadratic bond Hamiltonians; (iii) one-photon loss with rate $\kappa \geq 0$ and directional bond jumps proportional to $a_x - a_y$; and (iv) on-site two-photon loss $(\gamma/N)\mathcal{D}[a_x^2]$ with a fixed coefficient $\gamma > 0$. Every local term is supported in a set contained in some graph ball $B_{r_0}(x)$. The direction-dependent realization constructed in Theorem 3.11, for which $\gamma = 2\nu > 0$, belongs to this class.

Proposition 2.4. *Let r_0 be the interaction range in the standing semiclassical assumptions, fix $\zeta > 0$, and set*

$$(2.4) \quad w_x(u) = e^{-\zeta d_L(x,u)}, \quad \mathcal{N}_x^{(\zeta)} = \sum_{u \in \Lambda_L} w_x(u) n_u, \quad \Phi_{x,\theta} = e^{\theta \mathcal{N}_x^{(\zeta)}}.$$

For sufficiently small fixed $\theta > 0$ there is $C(\theta, \zeta) < \infty$, independent of L, N , the cutoff ratio c , and $M_N = \lfloor cN \rfloor$, such that every density matrix ρ on the truncated space satisfies

$$(2.5) \quad \rho(\mathcal{L}_{L,N}(\Phi_{x,\theta})) \leq C(\theta, \zeta) N \rho(\Phi_{x,\theta}).$$

The occupation-basis argument also yields a nearby-center form with a finite sum over $d_L(x, z) \leq r_0$.

Proof. The truncated number operator $n_u = a_{u,M_N}^* a_{u,M_N}$ obeys, including on the top cutoff state,

$$(2.6) \quad [n_u, a_{u,M_N}] = -a_{u,M_N}, \quad [n_u, a_{u,M_N}^*] = a_{u,M_N}^*.$$

$$(2.7) \quad e^{sn_u} a_{u,M_N} e^{-sn_u} = e^{-s} a_{u,M_N}, \quad e^{sn_u} a_{u,M_N}^* e^{-sn_u} = e^s a_{u,M_N}^*,$$

with constants independent of the cutoff. This is the point at which the weighted-number exponential observable differs from a plateau weight.

Write $\mathcal{L}_{L,N} = \sum_Z \mathcal{L}_Z$, where $\text{diam } Z \leq r_0$. Bounded geometry and exponential summability give

$$\sup_{L,x} \sum_Z \sum_{u \in Z} w_x(u) < \infty,$$

and weights within one interaction set are comparable by a factor depending only on r_0 and ζ .

Weighted monomial bounds. For a normal-ordered monomial $X_Z = \prod_{u \in Z} (a_u^*)^{p_u} a_u^{q_u}$ define $\Delta_x(X_Z) = \sum_{u \in Z} (p_u - q_u) w_x(u)$. Equation (2.7) yields

$$\Phi_{x,\theta} X_Z \Phi_{x,\theta}^{-1} = e^{\theta \Delta_x(X_Z)} X_Z, \quad |e^{\theta \Delta_x(X_Z)} - 1| \leq C_{r_0, \zeta} \theta \sum_{u \in Z} w_x(u)$$

for every monomial occurring in the generator. Reindexing occupation-basis matrix elements and using positivity in the form $|\rho_{\mathbf{m}, \mathbf{n}}| \leq (\rho_{\mathbf{m}, \mathbf{m}} \rho_{\mathbf{n}, \mathbf{n}})^{1/2}$ yields the following explicit bounds, where $u \sim v$ means that $\{u, v\}$ is the support of one finite-range two-site bond term of the generator and all constants are independent of the cutoff:

$$(2.8) \quad |\rho(i[a_u^2 + a_u^{*2}, \Phi_{x,\theta}])| \leq C\theta w_x(u) \rho((n_u + 1)\Phi_{x,\theta}),$$

$$(2.9) \quad |\rho(i[a_u^* a_v + a_v^* a_u, \Phi_{x,\theta}])| \leq C\theta (w_x(u) + w_x(v)) \rho((n_u + n_v + 2)\Phi_{x,\theta}),$$

$$(2.10) \quad |\rho(i[a_u^* a_v^* + a_u a_v, \Phi_{x,\theta}])| \leq C\theta (w_x(u) + w_x(v)) \rho((n_u + n_v + 2)\Phi_{x,\theta}),$$

$$(2.11) \quad [\rho(\mathcal{D}[a_u - a_v]^*(\Phi_{x,\theta}))]_+ \leq C\theta (w_x(u) + w_x(v)) \rho((n_u + n_v + 2)\Phi_{x,\theta}),$$

$$(2.12) \quad |\rho(i[\sqrt{N}(f_u a_u^* + \bar{f}_u a_u), \Phi_{x,\theta}])| \leq C\theta w_x(u) \rho(\sqrt{N(n_u + 1)} \Phi_{x,\theta}).$$

For the pairing term, the occupation-basis reindexing is as follows. Let $\mathbf{e}_u$ denote one quantum at site u , write

$$\Phi_{x,\theta} |\mathbf{n}\rangle = \phi_{\mathbf{n}} |\mathbf{n}\rangle, \quad \phi_{\mathbf{n}+\mathbf{e}_u+\mathbf{e}_v} = e^{\theta \Delta_{uv}} \phi_{\mathbf{n}}, \quad \Delta_{uv} = w_x(u) + w_x(v),$$

and set $X = a_u^* a_v^*$. The sum below is over occupations for which the shifted basis vector remains inside the cutoff. Positivity of the density matrix and $2\sqrt{ab} \leq a + b$ give

$$\begin{aligned} |\rho(i[X + X^*, \Phi_{x,\theta}])| &\leq 2(e^{\theta \Delta_{uv}} - 1) \sum_{\mathbf{n}}' \sqrt{(n_u + 1)(n_v + 1)} \phi_{\mathbf{n}} |\rho_{\mathbf{n}, \mathbf{n}+\mathbf{e}_u+\mathbf{e}_v}| \\ &\leq (e^{\theta \Delta_{uv}} - 1) \sum_{\mathbf{n}}' \sqrt{(n_u + 1)(n_v + 1)} \phi_{\mathbf{n}} (\rho_{\mathbf{n}, \mathbf{n}} + \rho_{\mathbf{n}+\mathbf{e}_u+\mathbf{e}_v, \mathbf{n}+\mathbf{e}_u+\mathbf{e}_v}). \end{aligned}$$

In the second diagonal sum put $\mathbf{m} = \mathbf{n} + \mathbf{e}_u + \mathbf{e}_v$; then $\phi_{\mathbf{m}-\mathbf{e}_u-\mathbf{e}_v} = e^{-\theta \Delta_{uv}} \phi_{\mathbf{m}}$ and $2\sqrt{m_u m_v} \leq m_u + m_v$. Together with $2\sqrt{(n_u + 1)(n_v + 1)} \leq n_u + n_v + 2$ and $e^{\theta \Delta_{uv}} - 1 \leq C\theta (w_x(u) + w_x(v))$, this is (2.10). The

hopping estimate follows from an analogous one-quantum reindexing. Expanding the bond dissipator into its two diagonal loss terms and two cross terms gives (2.11); the diagonal one-photon contributions are nonpositive. The drive estimate follows from Cauchy–Schwarz. Let $\mathcal{L}_{L,N}^{\text{rest}}$ denote the sum of the on-site quadratic-Hamiltonian, quadratic bond-Hamiltonian, bond-dissipator, and coherent-drive contributions to $\mathcal{L}_{L,N}$. Summing (2.8)–(2.12) over their supports gives

$$(2.13) \quad \left[\rho \left(\mathcal{L}_{L,N}^{\text{rest}}(\Phi_{x,\theta}) \right) \right]_+ \leq C\theta \sum_u w_x(u) \rho \left([n_u + 1 + \sqrt{N(n_u + 1)}] \Phi_{x,\theta} \right).$$

The on-site one-photon-loss contribution is nonpositive. The two-photon-loss contribution has the exact diagonal identity

$$(2.14) \quad \frac{\gamma}{N} \mathcal{D}[a_u^2]^*(\Phi_{x,\theta}) = \frac{\gamma}{N} n_u(n_u - 1) (e^{-2\theta w_x(u)} - 1) \Phi_{x,\theta}.$$

After decreasing θ if necessary, $1 - e^{-2\theta w} \geq c_0 \theta w$ for $0 \leq w \leq 1$, and hence

$$\rho \left(\frac{\gamma}{N} \mathcal{D}[a_u^2]^*(\Phi_{x,\theta}) \right) \leq -\frac{c_0 \gamma \theta}{N} w_x(u) \rho(n_u(n_u - 1) \Phi_{x,\theta}).$$

The scalar estimate

$$(2.15) \quad C(n+1) + C\sqrt{N(n+1)} - \frac{c_0 \gamma}{N} n(n-1) \leq C'N, \quad n \geq 0,$$

absorbs every high-occupation factor in (2.13) at its own site. Summing over finite-range terms and using the uniform exponential-weight sum proves (2.5). $\square$

2.2.1. Finite-time cutoff control. A *volume-uniform compact amplitude class* is a family $\mathcal{K} = \{\mathcal{K}_L\}_{L \geq 1}$ in which each $\mathcal{K}_L \subset \ell^\infty(\Lambda_L; \mathbb{C})$ is compact in the sup norm and

$$\sup_{L \geq 1} \sup_{\alpha \in \mathcal{K}_L} \sup_{x \in \Lambda_L} |\alpha_x| < \infty.$$

All occurrences of a compact set $\mathcal{K}$ below have this meaning; for a fixed period cell it is obtained by periodic repetition on compatible tori.

Static coherent-state estimates require only $c > E$. For propagation on a prescribed time interval, the cutoff ratio is chosen after the horizon:

$$\forall T < \infty \quad \exists c_T < \infty \quad \forall c \geq c_T.$$

Lemma 2.5. *Fix $T < \infty$ and a volume-uniform compact amplitude class $\mathcal{K}$. Let E_0 bound the initial occupations $|\alpha_x(0)|^2$ uniformly in volume. There exists $c_T = c_T(\mathcal{K}) < \infty$ such that, for every cutoff ratio $c \geq c_T$ and $M_N = \lfloor cN \rfloor$, projected coherent product initial data with amplitudes in $\mathcal{K}$ satisfy the following estimate. For every fixed buffer C_{buf} there are $\zeta, \theta, c_{\text{tail}} > 0$ and $C_{T,C_{\text{buf}}} < \infty$, independent of L, N , and every larger c , such that*

$$(2.16) \quad \sup_{0 \leq t \leq T} \sup_{x \in \Lambda_L} \rho_{L,N}(t) (\mathbf{1}_{\{n_x \geq M_N - C_{\text{buf}}\}}) \leq C_{T,C_{\text{buf}}} e^{-c_{\text{tail}} N}.$$

In particular this probability is $O(N^{-K_0})$ for every fixed K_0 .

Proof. Use the weighted-number exponential observable (2.4). At one site, the projected coherent occupation law is a Poisson random variable X of mean $N|\alpha|^2$ conditioned on $X \leq M_N$. Conditioning on this lower event stochastically decreases X , so for every $t \geq 0$,

$$\mathbb{E}[e^{tX} \mid X \leq M_N] \leq \mathbb{E}[e^{tX}].$$

The product structure yields the cutoff-independent initial estimate

$$(2.17) \quad \rho_0(\Phi_{x,\theta}) \leq \exp \left\{ NE_0 \sum_{y \in \Lambda_L} (e^{\theta e^{-\zeta d_L(x,y)}} - 1) \right\} \leq e^{Nm_0(\theta, \zeta)},$$

where m_0 is uniform in L . [Proposition 2.4](#) and Gronwall imply

$$\sup_{0 \leq t \leq T} \sup_x \rho_t(\Phi_{x,\theta}) \leq e^{Nm_T(\theta,\zeta)}, \quad m_T = m_0 + TC(\theta,\zeta).$$

On the boundary event, $\Phi_{x,\theta} \geq e^{\theta(M_N - C_{\text{buf}})}$. Markov's inequality and $M_N = \lfloor cN \rfloor$ give

$$(2.18) \quad \rho_t(n_x \geq M_N - C_{\text{buf}}) \leq C_{C_{\text{buf}},\theta} \exp\{-N[\theta c - m_T(\theta,\zeta)]\}.$$

Fix ζ and then a sufficiently small θ . Choose c_T so that

$$\delta_T := \theta c_T - m_T(\theta,\zeta) > 0, \quad c_{\text{tail}} := \frac{\delta_T}{2}.$$

For every $c \geq c_T$, the exponent in (2.18) is at least $2c_{\text{tail}}N$; the floor and the fixed buffer are absorbed into the prefactor. Hence (2.16) holds with quantifier order $\forall T \exists c_T \forall c \geq c_T$. $\square$

2.2.2. Centered moments and first-moment convergence. For the evolved state define, before invoking any cubic closure,

$$(2.19) \quad \begin{aligned} \delta Q_{x,N}(t) &= Q_{x,N} - \rho_t(Q_{x,N}), & \delta P_{x,N}(t) &= P_{x,N} - \rho_t(P_{x,N}), \\ C_x(t) &= \rho_t(\delta Q_{x,N}(t)^2 + \delta P_{x,N}(t)^2). \end{aligned}$$

Lemma 2.6. *Under the hypotheses of [Lemma 2.5](#), fix a cutoff ratio $c \geq c_T$. Let $\mathcal{P}_x$ be a local polynomial of total degree at most three in the rescaled quadratures $Q_{y,N}, P_{y,N}$ with $y \in B_{r_0}(x)$. The expression $\mathcal{P}_x(\rho_t(Q), \rho_t(P))$ means that every $Q_{y,N}$ and $P_{y,N}$ in the polynomial is replaced by its expectation $\rho_t(Q_{y,N})$ and $\rho_t(P_{y,N})$, respectively. If the polynomial coefficients and first moments remain bounded on $[0, T]$, then*

$$(2.20) \quad |\rho_t(\mathcal{P}_x) - \mathcal{P}_x(\rho_t(Q), \rho_t(P))| \leq C_{\mathcal{P},T,c} \left(\max_{y \in B_{r_0}(x)} C_y(t) + N^{-1} \right) + C_{\mathcal{P},T,c} e^{-c_{\text{bd}}N},$$

uniformly in L, N, x , and $0 \leq t \leq T$, for some $c_{\text{bd}} > 0$.

Proof. Normal order $\mathcal{P}_x$ and write each rescaled quadrature as its mean plus the centered operator in (2.19). Terms with one centered factor vanish. Each ordering change contributes one rescaled canonical commutator of size $O(N^{-1})$, apart from a boundary projector controlled by [Lemma 2.5](#).

We make the noncommuting bulk insertion explicit. Write a normal-ordered monomial as

$$M = \prod_{y \in B_{r_0}(x)} M_y, \quad \deg M_y = d_y, \quad d_M := \sum_{y \in B_{r_0}(x)} d_y \leq 3,$$

where factors at distinct sites commute, and set $\Pi_{y,m} = \mathbf{1}_{\{n_y \leq m\}}$. Repeated use of $a_y \Pi_{y,m} = \Pi_{y,m-1} a_y \Pi_{y,m}$ and $a_y^* \Pi_{y,m} = \Pi_{y,m+1} a_y^* \Pi_{y,m}$ yields the sitewise shift identity

$$(2.21) \quad M_y \Pi_{y,M_N - C_{\text{buf}}} = \Pi_{y,M_N - C_{\text{buf}} + d_y} M_y \Pi_{y,M_N - C_{\text{buf}}}.$$

Choose once and for all $C_{\text{buf}} > 2d_M + 2r_0 + 2$, and define

$$\Pi_x^- := \prod_{y \in B_{r_0}(x)} \mathbf{1}_{\{n_y \leq M_N - C_{\text{buf}}\}}, \quad \Pi_x^+ := \prod_{y \in B_{r_0}(x)} \mathbf{1}_{\{n_y \leq M_N - C_{\text{buf}} + d_y\}}.$$

Multiplying the sitewise identities gives

$$(2.22) \quad M \Pi_x^- = \Pi_x^+ M \Pi_x^-.$$

Every degree- d_M word maps $\text{Ran } \Pi_x^-$ into $\text{Ran } \Pi_x^+$ without commuting a projector through the polynomial. For a product XYZ of at most three centered factors, insert $I = \Pi_x^- + (I - \Pi_x^-)$ on the right. In the term ending in Π_x^- , insert $I = \Pi_x^+ + (I - \Pi_x^+)$ on the left:

$$\rho_t(XYZ) = \rho_t(\Pi_x^+ XYZ \Pi_x^-) + \rho_t((I - \Pi_x^+) XYZ \Pi_x^-) + \rho_t(XYZ(I - \Pi_x^-)).$$

No projector is commuted through the polynomial.

For $X, Y \in \{Q, P\}$, write $\delta X_y = X_{y,N} - \rho_t(X_{y,N})$ and $\delta Y_z = Y_{z,N} - \rho_t(Y_{z,N})$. Two centered factors based at one site are bounded by $C_y(t)$ and at different sites by noncommutative Cauchy–Schwarz:

$$(2.23) \quad |\rho_t(\delta X_y \delta Y_z)| \leq \rho_t(\delta X_y \delta X_y^*)^{1/2} \rho_t(\delta Y_z^* \delta Y_z)^{1/2} \leq C(C_y(t) + C_z(t) + N^{-1}).$$

For three centered factors, on $\text{Ran } \Pi_x^+$ one rescaled field has norm at most $C\sqrt{c}$, while the remaining two are controlled by (2.23). On the complement, a fixed-degree rescaled polynomial has norm at most $C_{\mathcal{P},c}$ and the union bound plus (2.18) gives an exponentially small probability. Cauchy–Schwarz turns every bulk–boundary cross term into $C_{\mathcal{P},T,c}e^{-c_{\text{bd}}N}$ with, for example, $c_{\text{bd}} = c_{\text{tail}}/2$. Summing the finitely many normal-ordered monomials proves (2.20). $\square$

Lemma 2.7. *Fix $T < \infty$ and a volume-uniform compact amplitude class $\mathcal{K}$. Choose $c \geq c_T(\mathcal{K})$ as in Lemma 2.5. For projected coherent product initial data with amplitudes in $\mathcal{K}$, the evolved finite- N states satisfy*

$$(2.24) \quad \sup_{0 \leq t \leq T} \sup_{x \in \Lambda_L} C_x(t) \leq C_{T,c} N^{-1},$$

where $C_{T,c}$ is independent of L and N .

Proof. Fix a site x and write

$$m_x(t) = \rho_t(a_x), \quad b_x = a_x - m_x(t), \quad v_x(t) = \rho_t(b_x^* b_x).$$

Away from the exponentially small cutoff boundary, $C_x = (2v_x + 1)/N + O(e^{-c_{\text{tail}}N})$. For an individual Heisenberg-generator contribution $\mathcal{G}$,

$$(2.25) \quad \left. \frac{d}{dt} v_x \right|_{\mathcal{G}} = \rho_t(\mathcal{G}(a_x^* a_x)) - 2 \text{Re}[\overline{m_x} \rho_t(\mathcal{G}(a_x))].$$

For one-photon loss, take $\mathcal{G} = \kappa \mathcal{D}[a_x]^*$. Since $\mathcal{D}[a_x]^*(a_x) = -a_x/2$ and $\mathcal{D}[a_x]^*(a_x^* a_x) = -a_x^* a_x$, we obtain

$$(2.26) \quad \dot{v}_x|_{\kappa \mathcal{D}[a_x]^*} = -\kappa v_x, \quad \dot{C}_x|_{\kappa \mathcal{D}[a_x]^*} = -\kappa C_x + \frac{\kappa}{N} + O(e^{-c_{\text{tail}}N}).$$

A coherent linear drive changes m_x , but its contribution to (2.25) cancels.

Every quadratic Hamiltonian gives a linear centered equation

$$\dot{b}_x|_H = \sum_{y \in B_{r_0}(x)} (U_{xy} b_y + W_{xy} b_y^*),$$

where $U_{xy}, W_{xy} \in \mathbb{C}$ vanish for $y \notin B_{r_0}(x)$, are uniformly bounded, and collect the onsite rotation, onsite squeezing, hopping, and bond-pairing coefficients. Thus

$$(2.27) \quad \dot{v}_x|_H = 2 \text{Re} \sum_y [U_{xy} \rho_t(b_x^* b_y) + W_{xy} \rho_t(b_x^* b_y^*)].$$

Noncommutative Cauchy–Schwarz gives

$$|\rho_t(b_x^* b_y)| \leq \sqrt{v_x v_y}, \quad |\rho_t(b_x^* b_y^*)| \leq \sqrt{v_x (v_y + 1)},$$

so the right side of (2.27) is bounded by $C \max_{z \in B_{r_0}(x)} v_z + C$.

For the unit-rate bond jump $L_{xy} = a_x - a_y$,

$$\mathcal{D}[L_{xy}]^* a_x = -\frac{1}{2}(a_x - a_y), \quad \mathcal{D}[L_{xy}]^*(a_x^* a_x) = -a_x^* a_x + \frac{1}{2}(a_y^* a_x + a_x^* a_y).$$

Equation (2.25) yields

$$(2.28) \quad \dot{v}_x|_{\mathcal{D}[a_x - a_y]^*} = -v_x + \text{Re} \rho_t(b_x^* b_y) \leq -\frac{1}{2}v_x + \frac{1}{2}v_y.$$

Summing the finitely many bonds produces the neighborhood maximum; the fixed commutator in $C_x = (2v_x + 1)/N$ contributes only C/N .

For the nonlinear channel,

$$\mathcal{D}[a_x^2]^* a_x = -a_x^* a_x^2, \quad \mathcal{D}[a_x^2]^* (a_x^* a_x) = -2a_x^{*2} a_x^2.$$

Substitution of $a_x = m_x + b_x$ into (2.25) gives

$$(2.29) \quad \begin{aligned} \dot{v}_x|_{(\gamma/N)\mathcal{D}[a_x^2]^*} = & -\frac{\gamma}{N} \left[4|m_x|^2 v_x + \bar{m}_x^2 \rho_t(b_x^2) + m_x^2 \rho_t(b_x^{*2}) \right. \\ & \left. + 3\bar{m}_x \rho_t(b_x^* b_x^2) + 3m_x \rho_t(b_x^{*2} b_x) + 2\rho_t(b_x^{*2} b_x^2) \right]. \end{aligned}$$

The anomalous moments satisfy $|\rho_t(b_x^2)| \leq \sqrt{v_x(v_x + 1)}$. The scale of m_x follows directly from the exponential-moment bound: for a constant m_T independent of N , Jensen's inequality gives

$$e^{\theta \rho_t(n_x)} \leq \rho_t(e^{\theta n_x}) \leq e^{Nm_T}, \quad \rho_t(n_x) \leq \frac{Nm_T}{\theta}, \quad |m_x|^2 \leq \rho_t(n_x).$$

Hence $|m_x| \leq C_T \sqrt{N}$. Young's inequality controls the cubic terms in (2.29) by the final nonpositive quartic term and lower moments. With

$$Q_x(t) = \frac{\gamma}{N^2} \rho_t(b_x^{*2} b_x^2) \geq 0,$$

we obtain

$$(2.30) \quad \left. \frac{d}{dt} C_x(t) \right|_{2\text{ph}} \leq C_c C_x(t) + \frac{C_c}{N} - Q_x(t).$$

Cubic centered moments are bounded by Lemma 2.6. Normal-ordering defects are $O(N^{-1})$. Each cutoff discrepancy contains an original-space boundary projector; the cutoff-tail estimate, together with a fixed-degree cutoff norm bound, makes these terms exponentially small in N . Combining the preceding estimates gives

$$\frac{d}{dt} C_x(t) \leq C_{T,c} \max_{z \in B_{r_0}(x)} C_z(t) + \frac{C_{T,c}}{N} + C_{T,c} e^{-c_{\text{tail}} N} - Q_x(t).$$

Gronwall's lemma, applied to the maximum over x after dropping the favorable term Q_x , and the $O(N^{-1})$ projected coherent initial covariance prove (2.24). $\square$

Proposition 2.8. *Fix $T < \infty$ and a volume-uniform compact amplitude class $\mathcal{K}$. Choose $c \geq c_T(\mathcal{K})$ as in Lemma 2.5, and let $E < c$ exceed the bound supplied by Lemma 2.10. For projected coherent product initial data with amplitudes in $\mathcal{K}$, the finite- N Heisenberg first moments satisfy, uniformly in $x \in \Lambda_L$ and with constants independent of L ,*

$$\sup_{0 \leq t \leq T} \sup_{x \in \Lambda_L} (|\rho_{L,N}(t)(Q_{x,N}) - q_x(t)| + |\rho_{L,N}(t)(P_{x,N}) - p_x(t)|) \leq C_T N^{-1}.$$

Here $(q_x(t), p_x(t))$ solves the deterministic reaction–transport equation with matching initial coherent amplitudes.

Proof. Write

$$e_x(t) = (\rho_{L,N}(t)(Q_{x,N}) - q_x(t), \rho_{L,N}(t)(P_{x,N}) - p_x(t)).$$

Let $\mathcal{F}_\lambda^{\Lambda_L}$ denote the full lattice vector field, including reaction and transport, and let $\mathcal{F}_{\lambda,q,x}^{\Lambda_L}$ and $\mathcal{F}_{\lambda,p,x}^{\Lambda_L}$ be its two components at x . A direct Heisenberg commutator calculation yields, for the evolved state,

$$\frac{d}{dt} \rho_t(Q_{x,N}) = \mathcal{F}_{\lambda,q,x}^{\Lambda_L}(\rho_t(Q), \rho_t(P)) + \text{Err}_{q,x}^{\text{fluc}}(t) + \text{Err}_{q,x}^{\text{cut}}(t) + \text{Err}_{q,x}^{\text{ord}}(t),$$

and similarly for $P_{x,N}$. Here Err^{fluc} collects centered-moment terms, Err^{cut} collects terms containing a cutoff-boundary projection, and Err^{ord} collects normal-ordering commutators. The components of $\mathcal{F}_\lambda^{\Lambda_L}$

are the coherent symbols identified by [Lemma 2.3](#); this identifies the deterministic drift, while the error estimate itself is taken on the evolved state. The local moment expansion in [Lemma 2.6](#) yields

$$|\text{Err}_{q,x}^{\text{fluc}}(t)| + |\text{Err}_{p,x}^{\text{fluc}}(t)| \leq C_T \max_{y \in B_{r_0}(x)} C_y(t),$$

the finite-time cutoff-tail lemma supplies exponentially small cutoff-boundary errors for the chosen $c \geq c_T(\mathcal{K})$, and normal-ordering commutators contribute $O(N^{-1})$. The cutoff error includes both the boundary commutator in $[a_{M_N}, a_{M_N}^*]$ and the difference between projected and unprojected coherent symbols: the former is controlled for evolved states by [Lemma 2.5](#), while the latter is exponentially small at $t = 0$ by [Lemma 2.2](#). Hence [Lemma 2.7](#) implies

$$\frac{d}{dt} \sup_x |e_x(t)| \leq C_T \sup_x |e_x(t)| + C_T N^{-1}.$$

The initial first-moment error is $O(N^{-1})$ by the coherent cutoff-tail estimate of [Lemma 2.2](#). Gronwall's lemma proves the displayed uniform first-moment estimate. The constants depend on T , the fixed local polynomial degree, and the compact classical sector, but not on L or N . $\square$

2.3. Turing assumptions and Bragg order. Fix a field dimension $m_{\text{op}} \in \mathbb{N}$ and write the local quantum order parameter as

$$O_{x,N} = (O_{x,N}^1, \dots, O_{x,N}^{m_{\text{op}}}).$$

Its rescaled first moments are collected in $u_x \in \mathbb{R}^{m_{\text{op}}}$. In the explicit realization below $m_{\text{op}} = 2$, $O_{x,N} = (Q_{x,N}, P_{x,N})$, and $u_x = (q_x, p_x)$. Let $\mathcal{R}_{\text{hop}} \subset \mathbb{Z}^d$ be a fixed finite displacement set, let $F_\lambda : \mathbb{R}^{m_{\text{op}}} \rightarrow \mathbb{R}^{m_{\text{op}}}$ be the local reaction field, and let $D_{\xi,\lambda}(u) \in \mathbb{R}^{m_{\text{op}} \times m_{\text{op}}}$ for $\xi \in \mathcal{R}_{\text{hop}}$. We suppose that the first moments close to

$$(2.31) \quad \dot{u}_x = \mathcal{F}_\lambda^{\Lambda_L}(u)_x := F_\lambda(u_x) + \sum_{\xi \in \mathcal{R}_{\text{hop}}} D_{\xi,\lambda}(u_x)(u_{x+\xi} - u_x),$$

where addition of lattice sites is understood modulo L . Thus $\mathcal{F}_\lambda^{\Lambda_L} : (\mathbb{R}^{m_{\text{op}}})^{\Lambda_L} \rightarrow (\mathbb{R}^{m_{\text{op}}})^{\Lambda_L}$ is the full lattice drift, as distinguished from the local reaction field F_λ .

We identify the Brillouin torus with $\mathbb{T}^d := (-\pi, \pi]^d$ and the momentum grid of Λ_L with $\Lambda_L^* := (2\pi/L)(\mathbb{Z}/L\mathbb{Z})^d \subset \mathbb{T}^d$. With the unitary discrete Fourier convention

$$\widehat{f}(k) = |\Lambda_L|^{-1/2} \sum_{x \in \Lambda_L} e^{-ik \cdot x} f_x, \quad k \in \Lambda_L^*,$$

let $u_{\text{hom},\lambda}$ denote the homogeneous stationary state and let $\widehat{A}_\lambda(k)$ denote the symbol of $D_u \mathcal{F}_\lambda^{\Lambda_L}(u_{\text{hom},\lambda})$. The limiting equation lies in the classical reaction–diffusion and semilinear-parabolic setting [\[24, 25\]](#).

Definition 2.9. A homogeneous stationary state is a Turing point at a nonempty compact critical set $K_* \subset \mathbb{T}^d \setminus \{0\}$ when:

- (T1) $\widehat{A}_0(k)$ has a simple real zero eigenvalue for every $k \in K_*$;
- (T2) the homogeneous mode $\widehat{A}_0(0)$ is strictly stable;
- (T3) there are an open neighborhood $U \supset K_*$ and a spectral gap $\delta_{\text{gap}} > 0$ such that

$$\sup_{k \notin U} \max \text{Re Spec } \widehat{A}_0(k) \leq -\delta_{\text{gap}},$$

while inside U the critical eigenvalue vanishes only on K_* and has a nondegenerate quadratic maximum in every direction normal to K_* . For a finite symmetry orbit, this last condition is understood in all momentum directions.

In the explicit design below, the determinant has a quadratic zero as a function of the scalar lattice symbol ω , although the zero eigenvalue of the matrix is simple.

2.3.1. *Reaction–transport equation.* We consider two-component cubic reaction–transport equations for $u_x = (q_x, p_x)$ of the form

$$(2.32) \quad \begin{aligned} \dot{q}_x &= a q_x + \Omega_\lambda p_x - \nu(q_x^2 + p_x^2) q_x + \sum_{j=1}^d D_{q,j} \Delta_j q_x, \\ \dot{p}_x &= -\Omega_\lambda q_x - b p_x - \nu(q_x^2 + p_x^2) p_x + \sum_{j=1}^d D_{p,j} \Delta_j p_x, \end{aligned}$$

with $b > a > 0$, $D_{q,j}, D_{p,j} > 0$, and $\nu > 0$. We assume that $\lambda \mapsto \Omega_\lambda > 0$ is real analytic near 0. For the stripe, $(D_{q,1}, D_{p,1}) = (D_q, D_p)$ and $D_{q,j} = D_{p,j} = D_y$ for $j \geq 2$. The isotropic spot and labyrinth examples use $D_{q,j} = D_q$ and $D_{p,j} = D_p$ in every direction.

Lemma 2.10. *Every solution of (2.32) with uniformly bounded initial data is global. If*

$$M(t) := \max_{x \in \Lambda_L} (q_x(t)^2 + p_x(t)^2), \quad C_{\text{lin}} := 2a + 2 \sum_{j=1}^d \max\{D_{q,j}, D_{p,j}\},$$

then its upper Dini derivative satisfies

$$D^+ M(t) \leq C_{\text{lin}} M(t) - 2\nu M(t)^2.$$

Scalar comparison yields

$$M(t) \leq \max \left\{ M(0), \frac{C_{\text{lin}}}{2\nu} \right\}$$

for all $t \geq 0$, uniformly in L .

Proof. At a site where $q_x^2 + p_x^2 = M(t)$, the Ω_λ terms cancel. Using $2uv \leq u^2 + v^2$ for the two neighbors gives

$$2q_x \Delta_j q_x \leq 2(M(t) - q_x^2) = 2p_x^2, \quad 2p_x \Delta_j p_x \leq 2(M(t) - p_x^2) = 2q_x^2.$$

Hence

$$2D_{q,j} q_x \Delta_j q_x + 2D_{p,j} p_x \Delta_j p_x \leq 2 \max\{D_{q,j}, D_{p,j}\} M(t).$$

Dropping the favorable term $-2b p_x^2$ leaves a logistic differential inequality. Scalar comparison yields the displayed bound and rules out finite-time blow-up. $\square$

For a commensurate fixed-space reduction, let C be the chosen finite period cell, write $\mathcal{F}_\lambda^C$ for the periodic restriction of (2.31), and let $\mathcal{X}_{\text{fix}} \subset (\mathbb{R}^{m_{\text{op}}})^C$ be a real symmetry or isotropy fixed subspace with its normalized Euclidean inner product. Set

$$\mathcal{A}_* := D_u \mathcal{F}_0^C(u_{\text{hom},0})|_{\mathcal{X}_{\text{fix}}}.$$

Whenever $\ker \mathcal{A}_*$ is one-dimensional, choose a direct sum $\mathcal{X}_{\text{fix}} = \ker \mathcal{A}_* \oplus \mathcal{X}_{\text{ran}}$ and denote the associated projections by $\Pi_{\ker}$ and Π_{ran} .

Assumption 2.11. *Choose $k_* \in K_*$. At the nonzero critical set K_* , assume:*

- (A1) *the homogeneous state is a Turing point in the sense of Definition 2.9;*
- (A2) *the operator $\mathcal{A}_*$ has one-dimensional kernel and $\mathcal{A}_*|_{\mathcal{X}_{\text{ran}}}$ is invertible;*
- (A3) *the parameter λ unfolds the simple critical eigenvalue with positive crossing slope in that fixed space.*

When the critical set is a continuous shell, the fixed space in (A2) is chosen before the one-dimensional reduction. For an equation of the form (2.32), Lemma 3.2 gives supercritical saturation.

Choose right and left critical vectors $r, \ell \in \mathbb{R}^{m_{\text{op}}}$ with the normalization

$$\widehat{A}_0(k_*)r = 0, \quad \ell^\top \widehat{A}_0(k_*) = 0, \quad \ell^\top r = 1.$$

The crossing slope used below is the normalization-independent quantity

$$(2.33) \quad \chi_* := \frac{\ell^\top (\partial_\lambda \widehat{A}_\lambda(k_*)|_{\lambda=0})r}{\ell^\top r} > 0;$$

with the preceding normalization, the denominator equals one.

Assumption 2.12. *For the Bragg estimate, suppose that the reaction–transport equation has the form (2.32), satisfies the Turing hypotheses in Assumption 2.11, and has critical kernel*

$$\text{span}\{r\psi_*\}$$

in the chosen real symmetry-fixed space. Define the period lattice, its finite translation group, and its dual by

$$\Gamma_C := \{\gamma \in \mathbb{Z}^d : \psi_*(x + \gamma) = \psi_*(x) \text{ for every } x\}, \quad G_C := \mathbb{Z}^d / \Gamma_C, \\ G_C^* := \{k \in \mathbb{T}^d : e^{ik \cdot \gamma} = 1 \text{ for every } \gamma \in \Gamma_C\}.$$

We further assume:

- (G1) Γ_C has finite index, and a fixed fundamental period cell C tiles a sequence Λ_{L_j} of compatible tori with $L_j \rightarrow \infty$;
- (G2) there is a nonzero momentum $k \in K_* \cap G_C^*$ for which

$$\widehat{\psi}_*(k) = |C|^{-1} \sum_{x \in C} e^{-ik \cdot x} \psi_*(x)$$

is nonzero;

- (G3) there is a local pattern covector $v_{\text{pat}} \in (\mathbb{R}^{m_{\text{op}}})^*$ satisfying $v_{\text{pat}} \cdot r \neq 0$.

Let $u_{\lambda,0}$ denote a chosen fixed-space branch representative supplied by Theorem 3.3. Let $H_C \leq G_C$ be its stabilizer and let $\Theta := G_C/H_C$ be its finite translation orbit. For $\tau \in \Theta$, represented by any element of G_C , set

$$(2.34) \quad \psi_{*,\tau}(x) := \psi_*(x - \tau), \quad u_{\lambda,\tau}(x) := u_{\lambda,0}(x - \tau).$$

These definitions are independent of the chosen coset representative because H_C stabilizes the branch.

Theorem 2.13 (Lindblad realization and Bragg order). *Suppose the Bragg assumptions in Assumption 2.12 hold. Then the reaction–transport equation admits a family of Lindblad realizations with finite-range couplings, noncommuting order parameters, and the prescribed first-moment limit. Let $u_{\lambda,\tau}$ be a translate of the nonlinear branch and initialize the microscopic system in the projected coherent product state associated with $u_{\lambda,\tau}$. On every bounded time interval $[0, T]$, there is a cutoff threshold $c_T < \infty$ such that, for each $c \geq c_T$, the quantum structure-factor lower bound in Corollary 3.6 persists under the Lindblad evolution. The ratio c controls the finite- N Fock cutoff and does not enter the limiting reaction–transport equation.*

Proof. The bifurcation theorem supplies the nonlinear branch and Lemma 3.4 its deterministic structure-factor lower bound. The local parameter map in Theorem 3.11 realizes the reaction–transport equation, and Lemma 3.5 transfers the bound to projected coherent product states. For a fixed T , choose $c \geq c_T$ from Lemma 2.5; the first-moment estimate Proposition 2.8 then yields Corollary 3.6. $\square$

For finite L, N , the local order parameters are $O_{x,N} = (Q_{x,N}, P_{x,N})$, with $N[Q_{x,N}, P_{x,N}] \rightarrow i$ on bounded coherent-state sectors. The right critical eigenvector determines the local quadrature combination carrying the emerging pattern, whereas the left critical eigenvector determines the soft Heisenberg observable; in this paper it is

$$O_x^{\text{soft}} = \ell \cdot O_{x,N}, \quad \widehat{O}^{\text{soft}}(k_*) = |\Lambda_L|^{-1/2} \sum_x e^{-ik_* \cdot x} O_x^{\text{soft}}.$$

3. CONSTRUCTION AND STABILITY OF A COMMENSURATE TURING STRIPE

3.1. Spectral design and commensurate branches.

Theorem 3.1. *Let*

$$J_{\text{reac}} = \begin{pmatrix} a & \Omega \\ -\Omega & -b \end{pmatrix}, \quad b > a > 0,$$

and choose one lattice direction with transport coefficients $D_q, D_p > 0$. Assume

$$0 < \omega_* := \frac{aD_p - bD_q}{2D_qD_p} < 4, \quad \Omega^2 = ab + \frac{(aD_p - bD_q)^2}{4D_qD_p}.$$

Then the linearized system has a finite-wave-number Turing point at $2(1 - \cos k_) = \omega_*$ along that direction, and its determinant has a quadratic zero in the scalar symbol ω . The finite- k instability disappears when $D_q = D_p$.*

Proof. For $\omega = 2(1 - \cos k)$,

$$\det(J_{\text{reac}} - \omega \text{diag}(D_q, D_p)) = D_qD_p\omega^2 + (bD_q - aD_p)\omega + (\Omega^2 - ab).$$

The chosen value of Ω makes the discriminant zero and places the vertex at ω_* . The trace is negative there, so the zero eigenvalue is simple even though the determinant has second-order contact as a function of ω . Moreover $\text{tr } J_{\text{reac}} = a - b < 0$ and $\det J_{\text{reac}} = \Omega^2 - ab = (aD_p - bD_q)^2 / (4D_qD_p) > 0$, so the homogeneous reaction matrix is stable. The condition $0 < \omega_* < 4$ gives a lattice wave number. If $D_q = D_p = D$, then $J_{\text{reac}} - \omega DI$ shifts every eigenvalue of J_{reac} to the left, and scalar transport cannot destabilize the stable matrix J_{reac} at finite k . $\square$

If the pair (D_q, D_p) acts in every direction, the symbol depends on $\omega_{\text{tot}}(k) = \sum_j 2(1 - \cos k_j)$ and the critical set is the isotropic shell $\omega_{\text{tot}}(k) = \omega_*$. If differential transport acts only in direction 1 while the remaining directions have common scalar transport, the transverse scalar terms shift those modes to the left and isolate the longitudinal stripe orbit.

On the fixed period cell C , use the normalized Euclidean inner product and average

$$\langle f, g \rangle_C := \frac{1}{|C|} \sum_{x \in C} f(x) \cdot g(x), \quad \langle f \rangle_C := \frac{1}{|C|} \sum_{x \in C} f(x),$$

where the dot is omitted for scalar fields.

Lemma 3.2. *Assume that the critical kernel in the chosen real symmetry-fixed space is*

$$\text{span}\{r\psi_*\},$$

where $r, \ell \in \mathbb{R}^2$, $\ell^\top r = 1$, and ψ_ is a nonzero real critical spatial eigenfunction. If the crossing slope is $\chi_* > 0$, then the reduced equation has the form*

$$0 = B \left(\chi_* \lambda - c_3 B^2 + O(\lambda^2 + \lambda B^2 + B^4) \right),$$

where

$$c_3 = \nu |r|^2 \frac{\langle \psi_*, \psi_*^3 \rangle_C}{\langle \psi_*, \psi_* \rangle_C} = \nu |r|^2 \frac{\langle \psi_*^4 \rangle_C}{\langle \psi_*^2 \rangle_C} > 0.$$

For a single complex Fourier pair normalized as

$$u_{\text{crit}} = B r e^{ik_* \cdot x} + \bar{B} r e^{-ik_* \cdot x},$$

the resonant complex amplitude equation instead has cubic term $-3\nu |r|^2 B |B|^2$, so this specialization is $c_3 = 3\nu |r|^2$ in the complex Fourier-coefficient convention.

Proof. The nonlinearity has no quadratic part, so the range correction enters the critical projection only at the displayed higher orders. For a real fixed-space critical component $Br\psi_*$,

$$-v|Br\psi_*|^2(Br\psi_*) = -vB^3|r|^2r\psi_*^3.$$

Projection onto the left critical vector and spatial eigenfunction, with $\ell^\top r = 1$, produces the displayed overlap coefficient. Positivity follows because ψ_* is real and nonzero. For the complex pair, the coefficient of $e^{ik_* \cdot x}$ in $|u_{\text{crit}}|^2 u_{\text{crit}}$ is $3|B|^2 Br|r|^2$, which yields the Fourier-pair specialization. $\square$

Theorem 3.3. *Under the Turing assumptions in [Assumption 2.11](#), an equation of the form [\(2.32\)](#) has real-analytic maps*

$$s \mapsto (u(s), \lambda(s))$$

near $s = 0$, with

$$u(s) = s r \psi_* + O(s^3), \quad \lambda(s) = \frac{c_3}{\chi_*} s^2 + O(s^4),$$

where

$$c_3 = v|r|^2 \frac{\langle \psi_*^4 \rangle_C}{\langle \psi_*^2 \rangle_C} > 0.$$

The remainder is uniform in the period-cell ℓ^∞ norm. For $\lambda > 0$, inversion on either signed branch gives

$$u_\lambda^\pm = \pm \sqrt{\frac{\chi_*}{c_3}} \lambda^{1/2} r \psi_* + O(\lambda^{3/2}), \quad B_\lambda^\pm = \pm \sqrt{\frac{\chi_*}{c_3}} \lambda^{1/2} + O(\lambda^{3/2}).$$

Its leading Fourier support lies in K_* . If the equation satisfies [Assumption 2.12](#), every local pattern field with nonzero critical projection has the extensive semiclassical Bragg peak of [Lemma 3.4](#).

Proof. Write the solution as $s r \psi_* + z(s, \lambda)$ in the critical direction plus its range complement. Since $\mathcal{F}_\lambda^C(0) = 0$ and the nonlinearity has no quadratic term, the analytic range equation yields

$$z(s, \lambda) = O(|\lambda s| + |s|^3).$$

After division by the critical amplitude, [Lemma 3.2](#) becomes

$$0 = \chi_* \lambda - c_3 s^2 + O(\lambda^2 + \lambda s^2 + s^4).$$

The analytic implicit-function theorem solves this equation for $\lambda = \lambda(s)$ because $\chi_* > 0$. Substitution yields the branch, and the Bragg estimate follows from [Assumption 2.12](#) and [Lemma 3.4](#). $\square$

For the pattern covector v_{pat} , define the deterministic branch field by

$$O_{\lambda, \tau}^{\text{pat}}(x) := v_{\text{pat}} \cdot u_{\lambda, \tau}(x).$$

The $O(\lambda^{3/2})$ remainder in the branch expansion below is uniform in the period-cell ℓ^∞ norm, and hence on every period-compatible torus obtained by repetition of C . For every period-compatible torus Λ_L , define

$$\widehat{O}_{\lambda, \tau}^{\text{pat}}(k) = |\Lambda_L|^{-1/2} \sum_{x \in \Lambda_L} e^{-ik \cdot x} O_{\lambda, \tau}^{\text{pat}}(x),$$

$$S_\lambda^{\text{coh}}(k) = \frac{1}{|\Theta|} \sum_{\tau \in \Theta} \left| \widehat{O}_{\lambda, \tau}^{\text{pat}}(k) \right|^2.$$

The quantum operator structure factor is introduced in [Lemma 3.5](#). Because the momentum in (G2) belongs to G_C^* , $\tau \mapsto e^{-ik \cdot \tau}$ is a character of G_C . If $h \in H_C$, invariance of ψ_* under h gives

$$\widehat{\psi}_*(k) = e^{-ik \cdot h} \widehat{\psi}_*(k).$$

The coefficient is nonzero by (G2), so $e^{-ik \cdot h} = 1$. Therefore

$$\vartheta_k(\tau) := e^{-ik \cdot \tau}$$

is a well-defined nontrivial character of Θ , and

$$\widehat{\psi}_{*,\tau}(k) = \vartheta_k(\tau)\widehat{\psi}_*(k), \quad \frac{1}{|\Theta|} \sum_{\tau \in \Theta} \vartheta_k(\tau) = 0.$$

Lemma 3.4. *Assume the Bragg conditions and let $\text{Orb}(u_\lambda)$ denote the translation orbit*

$$u_{\lambda,\tau}(x) = B_\lambda r \psi_{*,\tau}(x) + O(\lambda^{3/2}), \quad \tau \in \Theta.$$

For the momentum k in (G2), set

$$c_{\text{Br}}(k) := \frac{\chi_*}{c_3} |\widehat{\psi}_*(k)|^2 > 0.$$

Then, for the covector v_{pat} in (G3) and every period-compatible torus Λ_{L_j} ,

$$\frac{S_\lambda^{\text{coh}}(k)}{|\Lambda_{L_j}|} \geq c_{\text{Br}}(k) |v_{\text{pat}} \cdot r|^2 \lambda - C \lambda^2.$$

For all sufficiently small $\lambda > 0$, the right side is at least $\frac{1}{2} c_{\text{Br}}(k) |v_{\text{pat}} \cdot r|^2 \lambda$. No off-diagonal translation-average assumption is required.

Proof. Fourier orthogonality at the fixed dual-lattice momentum k yields, uniformly on the translation orbit,

$$\widehat{O}_{\lambda,\tau}^{\text{pat}}(k) = |\Lambda_{L_j}|^{1/2} B_\lambda (v_{\text{pat}} \cdot r) \vartheta_k(\tau) \widehat{\psi}_*(k) + O(|\Lambda_{L_j}|^{1/2} \lambda^{3/2}).$$

Translation-orbit averaging changes no diagonal weight, because $|\vartheta_k| = 1$, and

$$\frac{S_\lambda^{\text{coh}}(k)}{|\Lambda_{L_j}|} = B_\lambda^2 |v_{\text{pat}} \cdot r|^2 |\widehat{\psi}_*(k)|^2 + O(\lambda^2).$$

Using $B_\lambda^2 = (\chi_*/c_3)\lambda + O(\lambda^2)$ identifies the coefficient. The absence of off-diagonal terms comes from evaluating one fixed Fourier momentum on a compatible torus. $\square$

Lemma 3.5. *Let $u_{\lambda,\tau}$ be the translation family in Lemma 3.4, write $\alpha_{\lambda,\tau,x} = (q_{\lambda,\tau,x} + ip_{\lambda,\tau,x})/\sqrt{2}$, and let*

$$\rho_{L,N,\lambda,\tau}^{\text{coh}} = |\psi_{\alpha_{\lambda,\tau}}^{(L,N)}\rangle \langle \psi_{\alpha_{\lambda,\tau}}^{(L,N)}|$$

be the associated projected coherent product state. Define the local quantum pattern observable and its Fourier mode by

$$O_{x,N}^{\text{pat}} := v_{\text{pat}} \cdot O_{x,N}, \quad \widehat{O}_{L,N}^{\text{pat}}(k) := |\Lambda_L|^{-1/2} \sum_{x \in \Lambda_L} e^{-ik \cdot x} O_{x,N}^{\text{pat}}.$$

$$S_{L,N,\lambda}^q(k) = \frac{1}{|\Theta|} \sum_{\tau \in \Theta} \rho_{L,N,\lambda,\tau}^{\text{coh}} \left(\widehat{O}_{L,N}^{\text{pat}}(k)^* \widehat{O}_{L,N}^{\text{pat}}(k) \right).$$

There are constants $C, c_0 > 0$, independent of j, L_j , and N , such that, for every compatible L_j ,

$$\frac{S_{L_j,N,\lambda}^q(k)}{|\Lambda_{L_j}|} \geq c_{\text{Br}}(k) |v_{\text{pat}} \cdot r|^2 \lambda - C \lambda^2 - C(N^{-1} + N e^{-c_0 N}).$$

Consequently,

$$\liminf_{j \rightarrow \infty} \liminf_{N \rightarrow \infty} \frac{S_{L_j,N,\lambda}^q(k)}{|\Lambda_{L_j}|} \geq c_{\text{Br}}(k) |v_{\text{pat}} \cdot r|^2 \lambda - C \lambda^2.$$

Proof. In this proof write

$$\widehat{O} := \widehat{O}_{L,N}^{\text{pat}}(k), \quad e_x := \rho_{L,N,\lambda,\tau}^{\text{coh}} \left(O_{x,N}^{\text{pat}} \right) - O_{\lambda,\tau}^{\text{pat}}(x).$$

Variance positivity yields, for every $\tau \in \Theta$,

$$\rho_{L,N,\lambda,\tau}^{\text{coh}}(\widehat{O}^* \widehat{O}) \geq \left| \rho_{L,N,\lambda,\tau}^{\text{coh}}(\widehat{O}) \right|^2.$$

By [Lemmas 2.2](#) and [2.3](#), the projected coherent expectation of each local pattern field equals the deterministic branch profile with a uniform error $O(N^{-1} + Ne^{-c_0N})$. The defined sitewise errors satisfy

$$\left| \frac{1}{|\Lambda_L|} \sum_x e_x e^{-ik \cdot x} \right| \leq \sup_x |e_x|,$$

so Fourier summation introduces no volume-dependent loss. Division by $|\Lambda_L|$, translation-orbit averaging, and [Lemma 3.4](#) prove the first inequality; the second follows by taking $N \rightarrow \infty$. $\square$

Corollary 3.6. *Under the hypotheses of [Lemma 3.5](#), let*

$$\rho_{L,N,\lambda,\tau}(t) = e^{t\mathcal{L}_{L,N}^*} \rho_{L,N,\lambda,\tau}^{\text{coh}}$$

be the actual Lindblad evolution and define

$$\mathbf{S}_{L,N,\lambda}^q(k, t) = \frac{1}{|\Theta|} \sum_{\tau \in \Theta} \rho_{L,N,\lambda,\tau}(t) \left(\widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k)^* \widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k) \right).$$

For every fixed $T < \infty$ there exists $c_T < \infty$ such that, for every cutoff ratio $c \geq c_T$, there are constants $C_T, c_0 > 0$, independent of L and N , such that, for every compatible L_j ,

$$\inf_{0 \leq t \leq T} \frac{\mathbf{S}_{L_j,N,\lambda}^q(k, t)}{|\Lambda_{L_j}|} \geq c_{\text{Br}}(k) |v_{\text{pat}} \cdot r|^2 \lambda - C\lambda^2 - C_T(N^{-1} + Ne^{-c_0N}).$$

The extensive lower bound in [Lemma 3.5](#) therefore holds after taking $N \rightarrow \infty$, uniformly for $0 \leq t \leq T$, and hence also along every compatible thermodynamic sequence.

Proof. For the compact finite branch orbit and prescribed T , choose $c \geq c_T$ from [Lemma 2.5](#). Since the deterministic branch is stationary, [Proposition 2.8](#) and [Lemmas 2.2](#) and [2.3](#) give

$$\sup_{0 \leq t \leq T, x} \left| \rho_{L,N,\lambda,\tau}(t) (\mathcal{O}_{x,N}^{\text{pat}}) - \mathcal{O}_{\lambda,\tau}^{\text{pat}}(x) \right| \leq C_T(N^{-1} + Ne^{-c_0N}).$$

Variance positivity at time t , followed by Fourier summation and [Lemma 3.4](#), gives the lower bound. $\square$

Corollary 3.7. *For the translation-orbit-averaged state*

$$\bar{\rho}_{L,N,\lambda}(t) = \frac{1}{|\Theta|} \sum_{\tau \in \Theta} \rho_{L,N,\lambda,\tau}(t),$$

define

$$\mathbf{S}_{\text{conn}}^q(k, t) = \bar{\rho}_{L,N,\lambda}(t) \left(\widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k)^* \widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k) \right) - \left| \bar{\rho}_{L,N,\lambda}(t) \left(\widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k) \right) \right|^2.$$

For the translation-covariant unseeded generator, the nontrivial character ϑ_k has zero orbit average. Hence

$$\bar{\rho}_{L,N,\lambda}(t) \left(\widehat{\mathcal{O}}_{L,N}^{\text{pat}}(k) \right) = 0, \quad \mathbf{S}_{\text{conn}}^q(k, t) = \mathbf{S}_{L,N,\lambda}^q(k, t).$$

The connected Bragg correlation is therefore extensive. This operator statement is not by itself an entanglement criterion; entanglement is analyzed separately through the fluctuation state.

Proof. Translation covariance carries translated branch states into translated evolved states. Averaging the nonzero-momentum one-point function over the nontrivial character ϑ_k gives zero, while linearity preserves the orbit-averaged two-point function. $\square$

For the explicit construction,

$$(3.1) \quad \dot{q}_x = q_x + \Omega_\lambda p_x - \nu(q_x^2 + p_x^2)q_x + D_{q,1}\Delta_1 q_x + D_y \Delta_2 q_x,$$

$$(3.2) \quad \dot{p}_x = -\Omega_\lambda q_x - 3p_x - \nu(q_x^2 + p_x^2)p_x + D_{p,1}\Delta_1 p_x + D_y \Delta_2 p_x,$$

where $D_y > 0$ is fixed arbitrarily; writing $D_q := D_{q,1}$ and $D_p := D_{p,1}$,

$$D_q = 1, \quad D_p = 3 + 2\sqrt{3}, \quad \Omega_\lambda = \sqrt{2\sqrt{3} - \lambda} > 0.$$

Then $k_* = (\pi/6, 0)$. Along the longitudinal period-12 cell, let

$$C_{12} := \mathbb{Z}/12\mathbb{Z}, \quad j \in C_{12},$$

and define the unit translation and the site- and bond-centered reflections by

$$T_{\text{lat}} j = j + 1, \quad \mathcal{R}_s j = -j, \quad \mathcal{R}_b j = -1 - j \pmod{12}.$$

They act on fields by pullback, $(g \cdot u)(j) = u(g^{-1}j)$.

Theorem 3.8 (Commensurate stripe branches). *At the Turing point $k_* = (\pi/6, 0)$, the homogeneous state of (3.1)–(3.2) undergoes a supercritical stationary bifurcation. The site- and bond-reflection-fixed spaces each have a one-dimensional critical kernel and an invertible range operator. They contain real-analytic branches parameterized by the signed critical amplitude s ,*

$$u^{\text{site}}(s, j) = 2s r \cos(k_* j) + O(s^3), \quad u^{\text{bond}}(s, j) = 2s r \cos(k_* j + \pi/12) + O(s^3),$$

$$\lambda(s) = 8\sqrt{3} \nu s^2 + O(s^4).$$

For $\lambda > 0$, the site- and bond-centered representatives and their lattice translates have leading harmonics

$$\begin{pmatrix} q_{\lambda, \mu}(x) \\ p_{\lambda, \mu}(x) \end{pmatrix} = 2B_\lambda r \cos(k_* \cdot x + \phi_0 + \mu\pi/6) + O(\lambda^{3/2}), \quad \mu \in C_{12},$$

where

$$r = \begin{pmatrix} r_q \\ r_p \end{pmatrix} = \begin{pmatrix} 1 \\ -\beta \end{pmatrix}, \quad \beta = \frac{\sqrt{3} - 1}{\sqrt{2\sqrt{3}}}, \quad B_\lambda = \sqrt{\frac{\lambda}{8\sqrt{3}\nu}} + O(\lambda^{3/2}).$$

Here $\phi_0 = 0$ for the site-centered representative and $\phi_0 = \pi/12$ for the bond-centered representative.

The translate indexed by $\mu \in C_{12}$ is fixed by $T_{\text{lat}}^\mu \mathcal{R}_s T_{\text{lat}}^{-\mu}$ or $T_{\text{lat}}^\mu \mathcal{R}_b T_{\text{lat}}^{-\mu}$, according to the representative.

Proof. For $k_2 = 0$,

$$\det \hat{A}_0(k_1, 0) = (1 - \omega)(-3 - (3 + 2\sqrt{3})\omega) + 2\sqrt{3} = (3 + 2\sqrt{3})(\omega - (2 - \sqrt{3}))^2.$$

At the homogeneous mode,

$$\hat{A}_0(0) = \begin{pmatrix} 1 & \sqrt{2\sqrt{3}} \\ -\sqrt{2\sqrt{3}} & -3 \end{pmatrix}, \quad \text{tr } \hat{A}_0(0) = -2, \quad \det \hat{A}_0(0) = 2\sqrt{3} - 3 > 0.$$

The zero wave number is strictly stable. Transverse scalar diffusion shifts every mode with $k_2 \neq 0$ strictly to the left, so the anisotropic critical orbit is $\{k_*, -k_*\}$. At k_* the trace is -4 , the crossing slope is $\chi_* = 1/4$, and right and left critical vectors may be chosen as

$$r = (1, -\beta)^\top, \quad \ell = \frac{(1, \beta)^\top}{1 - \beta^2}, \quad \ell^\top r = 1.$$

The period cell is equivariant under the dihedral group D_{12} generated by T_{lat} and the two reflections defined above. In $\text{Fix}(\mathcal{R}_s)$ the critical eigenspace is generated by $r \cos(k_* j)$; in $\text{Fix}(\mathcal{R}_b)$ it is generated by $r \cos(k_*(j + 1/2))$. The corresponding sine mode is odd and is absent. All remaining period-12 Fourier modes, including $3k_* = (\pi/2, 0)$, are separated from zero, and the transverse modes remain strictly stable. The standard simple-eigenvalue and equivariant Lyapunov–Schmidt argument applies [26–29]; the range equation is solved analytically by the implicit-function theorem.

Writing

$$u(x) = Bre^{ik_* \cdot x} + \overline{B}re^{-ik_* \cdot x} + w, \quad w \perp \text{span}\{re^{ik_* \cdot x}, re^{-ik_* \cdot x}\},$$

where $\perp$ refers to the normalized period-cell ℓ^2 inner product, extended sesquilinearly to complex Fourier modes. The range correction satisfies $w = O(|B|^3 + \lambda|B|)$. Projection by the normalized left vector $\ell^\top$ gives the resonant cubic term

$$-3\nu(1 + \beta^2)|B|^2 B = -2\sqrt{3}\nu|B|^2 B.$$

For $p, q \in \mathbb{N}_0$, a monomial $B^p \bar{B}^q$ is C_{12} -equivariant precisely when $p - q \equiv 1 \pmod{12}$. Denote by $\sigma_{\text{lock}} \in \mathbb{R}$ the coefficient of the first phase-locking monomial. The reduced stationary equation is

$$0 = B \left(\frac{\lambda}{4} - 2\sqrt{3}\nu|B|^2 + O(\lambda^2 + \lambda|B|^2 + |B|^4) \right) + \sigma_{\text{lock}} \bar{B}^{11} + O(|B|^{13} + \lambda|B|^{11}).$$

The selection rule makes $\bar{B}^{11}$ the first symmetry-allowed locking term. On either real reflection-fixed subspace this yields

$$\lambda(s) = 8\sqrt{3}\nu s^2 + O(s^4), \quad B_\lambda = \sqrt{\frac{\lambda}{8\sqrt{3}\nu}} + O(\lambda^{3/2}).$$

Equivariance produces the translated representatives and their conjugate reflections. $\square$

Theorem 3.9 (Stability in the reflection-fixed spaces). *Let $\mathcal{X}_{\text{site}}$ and $\mathcal{X}_{\text{bond}}$ be the site- and bond-reflection fixed subspaces of the real period-12 first-moment phase space. For every sufficiently small $\lambda > 0$, the corresponding branch u_λ^{site} or u_λ^{bond} is locally asymptotically stable for the period-cell first-moment dynamics restricted to its fixed space. More precisely, there are $\lambda_0, c_*, g_* > 0$ such that for $0 < \lambda < \lambda_0$ the linearization at either branch has one center-originating eigenvalue*

$$(3.3) \quad \mu_{\text{rad}}(\lambda) = -\frac{\lambda}{2} + O(\lambda^2),$$

and every remaining fixed-space eigenvalue satisfies $\text{Re } \mu \leq -g_*$. In particular the fixed-space spectral abscissa is at most $-c_*\lambda$.

Proof. On either reflection-fixed space, the critical eigenspace at $\lambda = 0$ is one-dimensional and all range eigenvalues lie in $\{\text{Re } z \leq -g_0\}$ for some $g_0 > 0$; this is the spectral decomposition used in [Theorem 3.8](#). The center-manifold coordinate may be chosen as the signed real critical amplitude B . The projected coefficients of the dynamical reduced equation coincide with those of the stationary bifurcation equation,

$$(3.4) \quad \dot{B} = \frac{\lambda}{4}B - 2\sqrt{3}\nu B^3 + O(\lambda^2 B + \lambda B^3 + B^5) + O(B^{11}).$$

At the nonzero branch, $B_\lambda^2 = \lambda/(8\sqrt{3}\nu) + O(\lambda^2)$. Differentiating (3.4) at $B = B_\lambda$ gives

$$\partial_B \dot{B}|_{B_\lambda} = \frac{\lambda}{4} - 6\sqrt{3}\nu B_\lambda^2 + O(\lambda^2) = -\frac{\lambda}{2} + O(\lambda^2),$$

which is negative for sufficiently small λ . The range spectrum remains in $\{\text{Re } z \leq -g_0/2\}$ by finite-dimensional spectral continuity and the analytic range correction. The standard linearized-stability theorem for a smooth finite-dimensional ODE then gives local asymptotic stability. No claim is made here about stability in the full period-cell space, where the commensurate phase-locking eigenvalue is of order λ^5 and its sign depends on the locking coefficient. $\square$

Remark 3.10. Since $k_* = \pi/6$, the lattice branch has period 12 and only a discrete translation orbit. The site- and bond-centered branches correspond to the two reflection fixed spaces constructed in [Theorem 3.8](#); the bond-centered branch $\phi_0 = \pi/12$ is the one used in the numerical section.

3.2. Lindblad realization. In this subsection we write

$$a_x := a_{x,M_N}, \quad Q_x := Q_{x,N}, \quad P_x := P_{x,N},$$

and suppress the cutoff subscripts to keep formulas readable. Let E_j be the nearest-neighbor bonds in direction e_j , and set

$$K_j = \frac{D_{q,j} + D_{p,j}}{2}, \quad K'_j = \frac{D_{q,j} - D_{p,j}}{2}.$$

For $h \geq 0$ and $\mu \in C_{12}$, define the optional coherent phase seed

$$H_{\text{seed}}^{h,\mu} = i\sqrt{N} \sum_x (h_x^{(\mu)} a_x^* - \overline{h_x^{(\mu)}} a_x), \quad h_x^{(\mu)} = \frac{h}{\sqrt{2}} (r_q + ir_p) \cos(k_* \cdot x + \phi_0 + \mu\pi/6).$$

The unseeded construction is obtained by setting $h = 0$; the optional seed is only a finite-volume phase selector for a discrete translate. The direction-dependent construction is

$$\begin{aligned} H_{\text{loc}} &= \sum_x \left[\Omega_\lambda a_x^* a_x + \frac{i\varepsilon}{2} ((a_x^*)^2 - a_x^2) \right], \\ H_{\text{sq}} &= -\frac{i}{2} \sum_{j=1}^d K'_j \sum_{\langle x,y \rangle \in E_j} [((a_x^* - a_y^*)^2) - (a_x - a_y)^2], \\ \mathcal{L}^*(\rho) &= -i[H_{\text{loc}} + H_{\text{sq}} + H_{\text{seed}}^{h,\mu}, \rho] + \kappa \sum_x \mathcal{D}[a_x] \rho + \frac{\gamma}{N} \sum_x \mathcal{D}[a_x^2] \rho \\ &\quad + 2 \sum_{j=1}^d K_j \sum_{\langle x,y \rangle \in E_j} \mathcal{D}[a_x - a_y] \rho. \end{aligned}$$

Theorem 3.11. *With*

$$\varepsilon = \frac{a+b}{2}, \quad \kappa = b-a, \quad K_j = \frac{D_{q,j} + D_{p,j}}{2}, \quad K'_j = \frac{D_{q,j} - D_{p,j}}{2}, \quad \gamma = 2\nu,$$

the unseeded generator realizes every equation of the form (2.32) in the local coherent-state topology, with an $O_{S_{L,N}(E)}(N^{-1})$ remainder uniformly on bounded coherent-state sets. Adding $H_{\text{seed}}^{h,\mu}$ adds precisely the deterministic forcing $hr \cos(k_ \cdot x + \phi_0 + \mu\pi/6)$ to the coherent first-moment equation. The stripe and isotropic specializations are obtained from the transport choices stated after (2.32).*

Proof. The local Hamiltonian and one-photon loss give

$$\dot{q} = (\varepsilon - \kappa/2)q + \Omega_\lambda p, \quad \dot{p} = -\Omega_\lambda q - (\varepsilon + \kappa/2)p.$$

For each j , dissipative hopping on E_j contributes $K_j \Delta_j$ to both quadratures, while the squeezing-hopping Hamiltonian contributes $K'_j \Delta_j$ to q and $-K'_j \Delta_j$ to p . Finally,

$$\frac{\gamma}{N} \mathcal{D}[a_x^2]^*(a_x) = -\frac{\gamma}{N} a_x^* a_x^2.$$

For

$$\alpha_x = N^{-1/2} \langle a_x \rangle = \frac{q_x + ip_x}{\sqrt{2}},$$

this gives $\dot{\alpha}_x = -\gamma |\alpha_x|^2 \alpha_x + O(N^{-1})$. The real cubic coefficient is $\gamma/2$, and $\gamma = 2\nu$ realizes the coefficient in (3.1)–(3.2). Combining the local terms gives

$$\begin{aligned} \mathcal{L}Q_x &= aQ_x + \Omega_\lambda P_x - \nu(Q_x^2 + P_x^2)Q_x + \sum_j D_{q,j} \Delta_j Q_x + O_{S_{L,N}(E)}(N^{-1}), \\ \mathcal{L}P_x &= -\Omega_\lambda Q_x - bP_x - \nu(Q_x^2 + P_x^2)P_x + \sum_j D_{p,j} \Delta_j P_x + O_{S_{L,N}(E)}(N^{-1}). \end{aligned}$$

All ordering errors are $O_{S_{L,N}(E)}(N^{-1})$ by [Lemma 2.3](#); the cutoff-tail contribution is exponentially small by [Lemma 2.2](#) and is absorbed in that coherent-state remainder. $\square$

Remark 3.12. The Lindblad construction also realizes positive radial odd-polynomial saturation. For a multiphoton order $r_{\text{ph}} \geq 2$, the local channel

$$\gamma_{r_{\text{ph}}} N^{1-r_{\text{ph}}} \mathcal{D}[a_x^{r_{\text{ph}}}]$$

contributes in the coherent limit

$$\dot{\alpha}_x = -\frac{r_{\text{ph}} \gamma_{r_{\text{ph}}}}{2} |\alpha_x|^{2r_{\text{ph}}-2} \alpha_x,$$

and hence a negative radial drift proportional to $-(q_x^2 + p_x^2)^{r_{\text{ph}}-1}(q_x, p_x)$. Finite sums of such channels therefore cover positive radial odd-polynomial reaction terms; the cubic model used below is the case $r_{\text{ph}} = 2$.

Lemma 3.13. *Let*

$$u_{\lambda,\mu}(x) = 2B_\lambda r \cos(k_* \cdot x + \phi_0 + \mu\pi/6) + O(\lambda^{3/2}), \quad \mu \in C_{12},$$

be the commensurate stripe branch, and let $O_x^{\text{pat}} = v_{\text{pat}} \cdot O_{x,N}$ with $v_{\text{pat}} \cdot r \neq 0$. The discrete phase-averaged weight satisfies

$$\frac{S_\lambda^{\text{coh}}(k_*)}{|\Lambda_L|} = B_\lambda^2 |v_{\text{pat}} \cdot r|^2 + O(\lambda^2) > 0$$

for all sufficiently small $\lambda > 0$ along period-compatible tori.

Proof. The $+k_*$ Fourier coefficient of $2B_\lambda \cos(k_* \cdot x + \phi_0 + \mu\pi/6)$ is $B_\lambda e^{i(\phi_0 + \mu\pi/6)}$, so the normalization of $S(k_*)$ gives the coefficient B_λ^2 rather than $2B_\lambda^2$. Since $B_\lambda = O(\lambda^{1/2})$, the cross term between the leading branch and the $O(\lambda^{3/2})$ branch correction contributes $O(\lambda^2)$ to the Bragg weight. Averaging over the twelve lattice translates gives

$$\frac{1}{12} \sum_{\mu=0}^{11} \cos(k_* \cdot x + \phi_0 + \mu\pi/6) \cos(k_* \cdot y + \phi_0 + \mu\pi/6) = \frac{1}{2} \cos(k_* \cdot (x - y)),$$

because the discrete sum of $e^{2i\mu\pi/6}$ vanishes. The discrete phase-averaged two-point function has leading term $2B_\lambda^2 |v_{\text{pat}} \cdot r|^2 \cos(k_* \cdot (x - y))$. Since $2k_* \not\equiv 0 \pmod{2\pi}$, Fourier orthogonality eliminates the nonzero $2k_*$ character. Substitution into the structure factor at k_* gives the displayed formula. This is the commensurate finite-orbit specialization of [Lemma 3.4](#). $\square$

The local pattern field is $O_x^{\text{pat}} = Q_x - \beta P_x$. The local soft field is $O_x^{\text{soft}} = Q_x + \beta P_x$; its Fourier component $\widehat{O}^{\text{soft}}(k_*)$ spans the left zero-eigenvalue mode of the homogeneous first-moment linearized Heisenberg generator at k_* .

4. PATTERN FORMATION AND QUANTUM FLUCTUATIONS

The first-moment dynamics displays three pattern classes. A weakly seeded stripe follows the analytic branch, while a broadband-noise run selects the same finite-wave-number band without a critical-mode seed. With isotropic transport, the instability produces spot and labyrinth regimes on a finite- k shell. Gaussian fluctuations are obtained by displacing the microscopic generator about the stationary stripe or along the two isotropic trajectories.

4.1. Numerical patterns and convergence. The simulations use periodic boundary conditions and the unitary discrete Fourier convention stated before [Definition 2.9](#). For the chosen parameter regime, let $A_{\text{lin}}(k) := \widehat{A}_\lambda(k)$ be the linear lattice symbol and let $G(u)_x = -\nu|u_x|^2 u_x$ be the onsite nonlinear remainder. The IMEX Euler update is

$$\widehat{u}^{n+1}(k) = [I - \Delta t_{\text{PDE}} A_{\text{lin}}(k)]^{-1} \left[\widehat{u}^n(k) + \Delta t_{\text{PDE}} \widehat{G(u^n)}(k) \right].$$

The baseline coherent-field step is $\Delta t_{\text{PDE}} := 0.05$. The stripe calculation uses

$$(\lambda, \nu, D_y, L, T) = (0.4, 4, 0.2, 192, 160).$$

The isotropic spot and labyrinth calculations use

$$(a, b, \Omega_\lambda, D_q, D_p, \nu, L) = (1, 3, 1.8, 0.6, 4.5, 4, 128),$$

with final times $T = 50$ and $T = 80$. Localized nuclei and weak noise select the Spot basin, while broadband noise produces the Labyrinth regime; in both cases the unstable band selects the wavelength.

The displayed pattern field is

$$O_{\text{stripe}}^{\text{pat}} = q - \beta p, \quad O_{\text{iso}}^{\text{pat}} = q - 0.40p.$$

For the isotropic parameters, the maximal linear growth rate occurs at $k \approx 0.649$, where the right eigenvector normalized by its q component is

$$r_{\text{max}} = (1, -0.365 \dots)^\top.$$

$v_{\text{iso}} = (1, -0.40)$ is a fixed visualization covector with nonzero overlap with the unstable eigenvector and is used unchanged for all isotropic runs. For either field, define its spatial mean by $\overline{O}^{\text{pat}} := |\Lambda_L|^{-1} \sum_{x \in \Lambda_L} O_x^{\text{pat}}$ and set

$$\widehat{O}^{\text{pat}}(k) := |\Lambda_L|^{-1/2} \sum_{x \in \Lambda_L} (O_x^{\text{pat}} - \overline{O}^{\text{pat}}) e^{-ik \cdot x}, \quad k_j = \frac{2\pi n_j}{L}, \quad n_j \in \mathbb{Z}/L\mathbb{Z}.$$

Radial bins are the half-open annuli of width $\Delta k = 2\pi/L$ centered at integer multiples of Δk ; $\overline{P}(\varrho)$ is the arithmetic mean of $|\widehat{O}^{\text{pat}}(k)|^2$ in the annulus centered at the radial coordinate ϱ . Define

$$k_{\text{dom}} = \arg \max_{\varrho > 0} \overline{P}(\varrho), \quad C_{\text{shell}} = \frac{\sum_{||k| - k_{\text{dom}}| \leq \Delta k} |\widehat{O}^{\text{pat}}(k)|^2}{\sum_{k \neq 0} |\widehat{O}^{\text{pat}}(k)|^2},$$

with ties resolved toward the smaller radius. Taking one Fourier spacing as the radial half-width makes C_{shell} resolution-normalized across grids. The reported stationary residual is

$$r_{\text{stat}} := \frac{\|\mathcal{F}_\lambda^{\Lambda_L}(u_{\text{final}})\|_2}{\|u_{\text{final}}\|_2}, \quad u_{\text{final}} = (q_{\text{final}}, p_{\text{final}}),$$

where $\mathcal{F}_\lambda^{\Lambda_L}$ is the full lattice drift in (2.31), specialized to the simulated regime. Refinement and control calculations preserve the selected finite-wave-number band. The stripe stationary residual is below 8.6×10^{-4} in the reported controls, and the Newton-continued period-cell profiles have infinity-norm residual below 10^{-8} .

At $\lambda = 0$ the maximal real eigenvalue of $\widehat{A}_\lambda(k_*)$ touches zero while the homogeneous mode remains stable; its derivative with respect to λ is $1/4$.

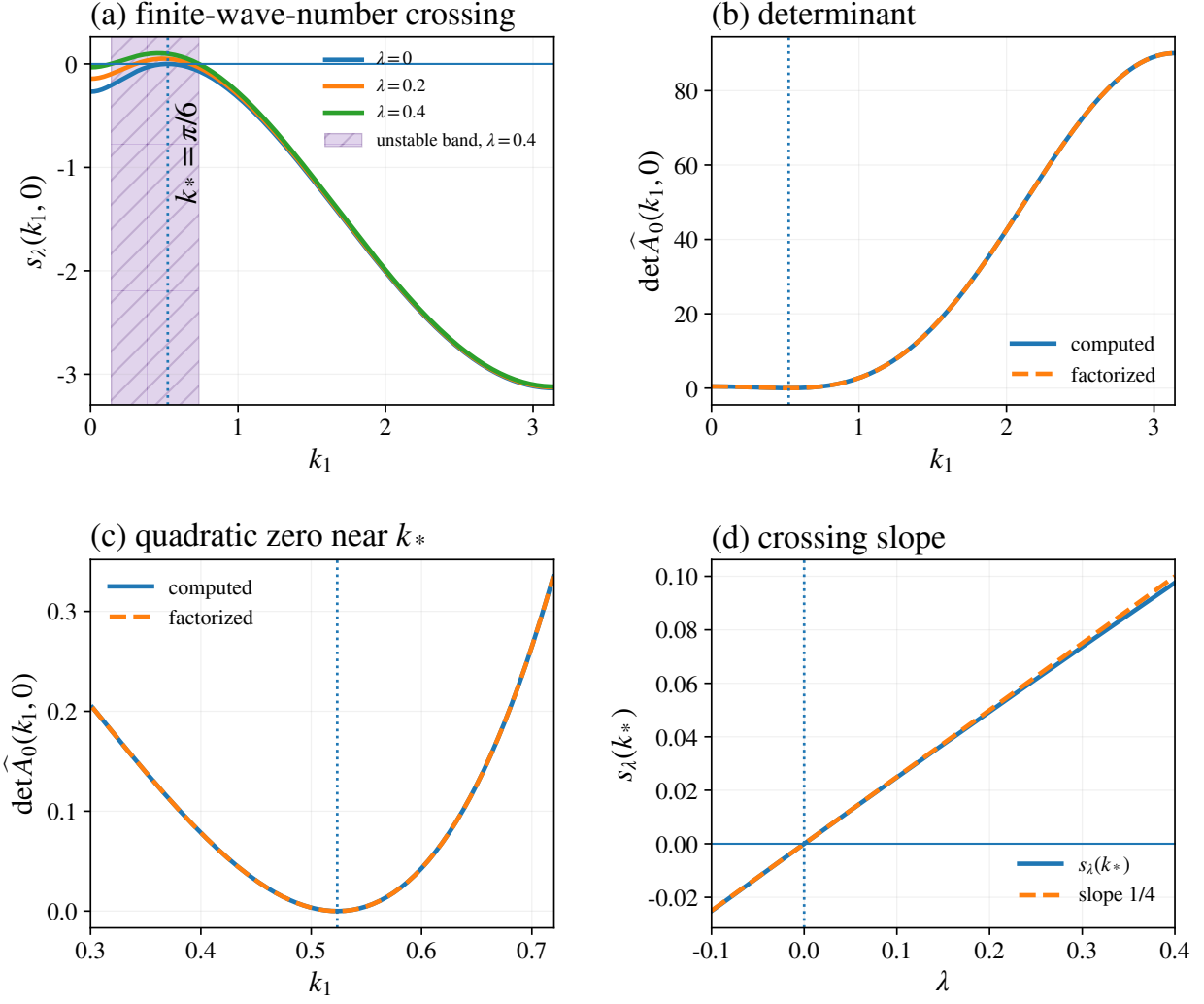


FIGURE 2. Finite wave-number Turing point. (a) The critical eigenvalue crosses zero at $k_* = \pi/6$ while the homogeneous mode remains stable; the shaded interval is the unstable band at $\lambda = 0.4$. (b,c) The computed determinant agrees with the exact factorization and has a quadratic zero at k_* . (d) The critical eigenvalue crosses with slope $1/4$.

The period-compatible grid represents $k_* = \pi/6$ without wave-number discretization error, and the measured $+k_*$ Fourier coefficient agrees with the leading branch prediction to about 0.2%.

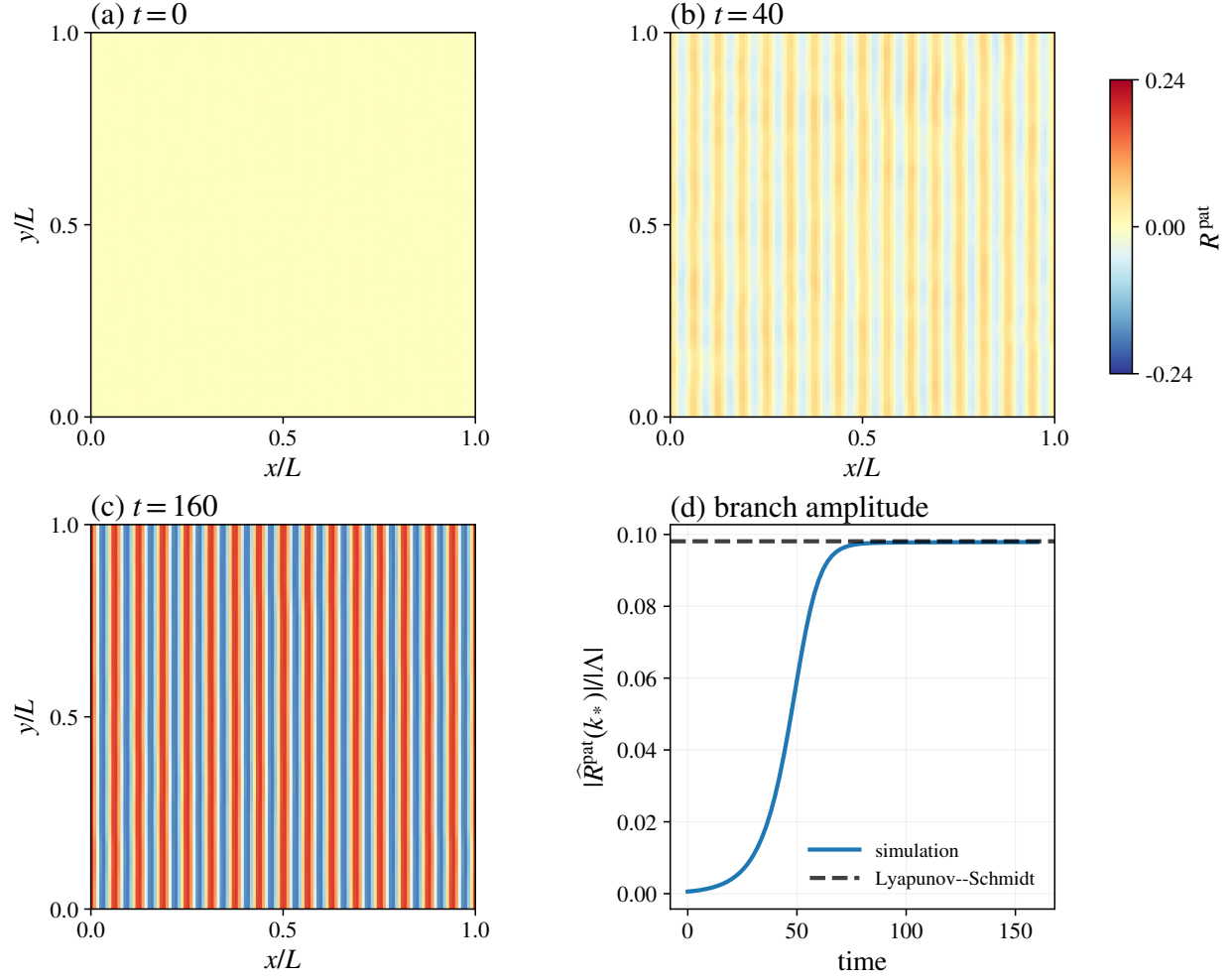


FIGURE 3. Evolution toward the commensurate stripe. (a–c) First-moment field at $t = 0$, 40, and 160. (d) The measured $+k_*$ Fourier amplitude approaches the Lyapunov–Schmidt prediction.

Across $0.05 \leq \lambda \leq 0.4$, the continued branch agrees with the leading Lyapunov–Schmidt amplitude law near threshold, the reflection-fixed spectral abscissa remains negative, and the period-cell NPT minimum remains at $k_*/\pi = 1/6$.

Regime	L	k_{dom}	C_{shell}
Stripe	192	0.52	> 0.99
Spot	128	0.59	0.56
Labyrinth	128	0.64	0.57

TABLE 1. Dominant wave number and shell concentration. The shell concentration uses one Fourier spacing as its radial half-width, as defined above.

The dominant power lies inside the corresponding linear unstable band in all three regimes.

4.2. Gaussian fluctuation spectra and mode entanglement. Let C be a finite period cell, with addition taken periodically, and define its directional bond set and periodic difference operators by

$$E_j(C) := \{\{x, x + e_j\} : x \in C\}, \quad (\Delta_j^C f)_x := f_{x+e_j} + f_{x-e_j} - 2f_x, \quad \mathbb{L}_j^C := -\Delta_j^C.$$

Let

$$u^\star = (q_x^\star, p_x^\star)_{x \in C}, \quad \alpha_x^\star = \frac{q_x^\star + ip_x^\star}{\sqrt{2}},$$

be a stationary periodic branch. Linearization of the local Lindblad generator produces a Gaussian dynamics on C . When its drift is Hurwitz, the stationary covariance is determined by an algebraic Lyapunov equation. The finite-time theorem below controls the approximation by the exact microscopic dynamics, and the homogeneous case is then solved in closed form.

Proposition 4.1. *Displace the microscopic modes according to*

$$a_x = \sqrt{N} \alpha_x^\star + b_x.$$

After cancellation of the $O(\sqrt{N})$ terms by the stationary branch equation, the $O(N^0)$ fluctuation generator is

$$(4.1) \quad \begin{aligned} \mathcal{L}_\star^{(2)*}(\rho) = & -i[H_\star^{(2)}, \rho] + \kappa \sum_{x \in C} \mathcal{D}[b_x] \rho + \sum_{x \in C} \mathcal{D}[2\sqrt{\gamma} \alpha_x^\star b_x] \rho \\ & + 2 \sum_{j=1}^d K_j \sum_{\langle x, y \rangle \in E_j(C)} \mathcal{D}[b_x - b_y] \rho, \end{aligned}$$

where

$$(4.2) \quad \begin{aligned} H_\star^{(2)} = & \sum_{x \in C} \left[\Omega_\lambda b_x^\star b_x + \frac{i}{2} \left\{ (\varepsilon - \gamma(\alpha_x^\star)^2) b_x^{*2} - (\varepsilon - \gamma(\overline{\alpha_x^\star})^2) b_x^2 \right\} \right] \\ & - \frac{i}{2} \sum_{j=1}^d K_j' \sum_{\langle x, y \rangle \in E_j(C)} \left[(b_x^\star - b_y^\star)^2 - (b_x - b_y)^2 \right]. \end{aligned}$$

Define the canonical fluctuation quadratures by

$$\delta q_x := \frac{b_x + b_x^\star}{\sqrt{2}}, \quad \delta p_x := \frac{b_x - b_x^\star}{i\sqrt{2}},$$

and collect them in

$$\xi := (\delta q_1, \delta p_1, \dots, \delta q_{|C|}, \delta p_{|C|})^\top.$$

Here and below $\langle \cdot \rangle$ denotes expectation in the Gaussian fluctuation state under discussion. The first-moment drift is

$$\frac{d}{dt} \langle \xi \rangle = A_{\text{per}} \langle \xi \rangle, \quad A_{\text{per}} := D_u \mathcal{F}_\lambda^C(u^\star),$$

where the site-diagonal and nearest-neighbor blocks are

$$(4.3) \quad (A_{\text{per}})_{xx} = \begin{pmatrix} a - \nu(3q_x^{*2} + p_x^{*2}) & \Omega_\lambda - 2\nu q_x^\star p_x^\star \\ -\Omega_\lambda - 2\nu q_x^\star p_x^\star & -b - \nu(q_x^{*2} + 3p_x^{*2}) \end{pmatrix} - 2 \sum_{j=1}^d \begin{pmatrix} D_{q,j} & 0 \\ 0 & D_{p,j} \end{pmatrix},$$

$$(4.4) \quad (A_{\text{per}})_{x, x \pm e_j} = \begin{pmatrix} D_{q,j} & 0 \\ 0 & D_{p,j} \end{pmatrix},$$

with periodic wrap on C . If $\mathbb{L}_j^C = -\Delta_j^C$ is the period-cell graph Laplacian, the symmetrized covariance

$$V_{mn} = \frac{1}{2} \langle \xi_m \xi_n + \xi_n \xi_m \rangle - \langle \xi_m \rangle \langle \xi_n \rangle$$

satisfies

$$(4.5) \quad \dot{V} = A_{\text{per}}V + VA_{\text{per}}^T + D_{\text{per}}, \quad D_{\text{per}} = \bigoplus_{x \in C} \left[\frac{\kappa}{2} + \gamma(q_x^{\star 2} + p_x^{\star 2}) \right] I_2 + \sum_{j=1}^d K_j \mathbb{L}_j^C \otimes I_2.$$

If A_{per} is Hurwitz, the stationary covariance is unique and is given by

$$V_{\text{per}} = \int_0^\infty e^{tA_{\text{per}}} D_{\text{per}} e^{tA_{\text{per}}^T} dt,$$

or equivalently by the algebraic Lyapunov equation

$$A_{\text{per}}V_{\text{per}} + V_{\text{per}}A_{\text{per}}^T + D_{\text{per}} = 0.$$

Proof. The shifted two-photon jump expands as

$$\sqrt{\gamma/N} a_x^2 = \sqrt{\gamma N} (\alpha_x^\star)^2 + 2\sqrt{\gamma} \alpha_x^\star b_x + \sqrt{\gamma/N} b_x^2.$$

For a scalar c and an operator X ,

$$\mathcal{D}[c\mathbf{1} + X]\rho = \mathcal{D}[X]\rho - i \left[\frac{i}{2} (\bar{c}X - cX^*), \rho \right].$$

The constant–linear cross terms, together with the displaced Hamiltonian and bond terms, are precisely the $O(\sqrt{N})$ stationary first-moment equation and vanish at $u^\star$. The constant–quadratic cross term produces

$$-\frac{i\gamma}{2} \left[(\alpha_x^\star)^2 b_x^{\star 2} - (\bar{\alpha}_x^\star)^2 b_x^2 \right],$$

while $2\sqrt{\gamma} \alpha_x^\star b_x$ is the surviving linear jump. The remaining quadratic-jump and linear–quadratic cross terms are lower order in $N^{-1/2}$. The other shifted jumps retain the linear parts displayed in (4.1), and the quadratic Hamiltonians retain the form in (4.2).

The first-moment equation of this Gaussian generator is the linearization of the semiclassical vector field, giving (4.3)–(4.4). One-photon loss contributes $\kappa I_2/2$ to the local diffusion, the linearized two-photon jump contributes $\gamma(q_x^{\star 2} + p_x^{\star 2})I_2$, and the jump $\sqrt{2K_j}(b_x - b_y)$ contributes K_j times the corresponding edge Laplacian in each quadrature. These contributions give D_{per} in (4.5). The standard moment equations for a Gaussian Lindblad generator then yield the Lyapunov equation and its integral solution when A_{per} is Hurwitz. $\square$

For comparison with the microscopic dynamics, let Λ be a fixed torus and let

$$u(t) = (q_x(t), p_x(t))_{x \in \Lambda}, \quad 0 \leq t \leq T,$$

be a bounded solution of the first-moment equation. Write $\alpha_x(t) = (q_x(t) + ip_x(t))/\sqrt{2}$. For a one-site amplitude α , define the normalized projected coherent state

$$|\psi_\alpha^{(N)}\rangle := \frac{\mathbf{P}_{\leq M_N} |\sqrt{N}\alpha\rangle}{\|\mathbf{P}_{\leq M_N} |\sqrt{N}\alpha\rangle\|}.$$

Let $\rho_{\Lambda, N}(t)$ be the unseeded finite-volume Lindblad evolution on Λ with initial state

$$(4.6) \quad \rho_{\Lambda, N}(0) = \bigotimes_{x \in \Lambda} |\psi_{\alpha_x(0)}^{(N)}\rangle \langle \psi_{\alpha_x(0)}^{(N)}|.$$

On one full-Fock mode we use the Weyl convention

$$W(z) := \exp(za^* - \bar{z}a), \quad W(z)^* a W(z) = a + z.$$

Let ι_N be the canonical embedding of the cutoff tensor product into the full bosonic Fock space, with operators and states extended by zero on its orthogonal complement. Set

$$W_N(t) := \bigotimes_{x \in \Lambda} W(\sqrt{N}\alpha_x(t)), \quad \bar{\rho}_{\Lambda, N}(t) := W_N(t)^* \iota_N(\rho_{\Lambda, N}(t)) W_N(t).$$

Thus

$$W_N(t)^*(a_x - \sqrt{N}\alpha_x(t))W_N(t) = b_x, \quad \mathfrak{R}_b := \sum_{x \in \Lambda} b_x^* b_x,$$

where the b_x are canonical full-Fock annihilation operators. All identities involving the unbounded full-Fock operators b_x, b_x^* and their polynomial Lindblad generators are first understood as quadratic-form identities on the finite-particle core

$$\mathcal{D}_{\text{fin}} = \text{span}_{\text{alg}} \{ |n_1, \dots, n_{|\Lambda|} \rangle : n_x \in \{0, 1, 2, \dots\} \}.$$

The moment bounds below justify taking expectations of these identities and extending them to the evolved states considered here. Up to a real scalar phase $\vartheta_N(t)I$, the derivative of the time-dependent Weyl displacement is

$$iW_N(t)^*\dot{W}_N(t) = i\sqrt{N} \sum_{x \in \Lambda} \left[\dot{\alpha}_x(t)b_x^* - \overline{\dot{\alpha}_x(t)}b_x \right] + \vartheta_N(t)I.$$

Let $H_{\Lambda,N}$ denote the Hamiltonian part of the finite-volume generator on Λ , and let $H_{\Lambda,N}^{\text{ext}} := \iota_N(H_{\Lambda,N})$ be its zero extension. The Hamiltonian in the moving frame is

$$(4.7) \quad H_{\text{mov}}(t) := W_N(t)^*H_{\Lambda,N}^{\text{ext}}W_N(t) - iW_N(t)^*\dot{W}_N(t).$$

The moving-frame Heisenberg generator $\tilde{\mathcal{L}}_{\Lambda,N}^{\text{fl}}(t)$ is defined on fluctuation polynomials by

$$(4.8) \quad \frac{d}{dt}\tilde{\rho}_{\Lambda,N}(t)(A) = \tilde{\rho}_{\Lambda,N}(t)(\tilde{\mathcal{L}}_{\Lambda,N}^{\text{fl}}(t)A).$$

Let $\mathcal{F}_{\lambda,\mathbb{C},x}^\Lambda(u) := (\mathcal{F}_{\lambda,q,x}^\Lambda(u) + i\mathcal{F}_{\lambda,p,x}^\Lambda(u))/\sqrt{2}$ be the complex form of the full first-moment drift. The total $O(\sqrt{N})$ linear Hamiltonian is therefore proportional to

$$i\sqrt{N} \sum_x \left[(\mathcal{F}_{\lambda,\mathbb{C},x}^\Lambda(u(t)) - \dot{\alpha}_x)b_x^* - \overline{(\mathcal{F}_{\lambda,\mathbb{C},x}^\Lambda(u(t)) - \dot{\alpha}_x)}b_x \right].$$

This linear term vanishes because $u(t)$ solves the first-moment equation. We denote by $\mathcal{L}_{u(t)}^{(2)}$ the surviving quadratic Heisenberg generator: its drift is $D_u\mathcal{F}_\lambda^\Lambda(u(t))$ and its diffusion is assembled from the linear parts of the displaced jumps. For a stationary branch $u(t) = u^*$, the displacement $W_N(t)$ is constant, and $\mathcal{L}_{u^*}^{(2)}$ is the Heisenberg adjoint of the Schrödinger-picture generator $\mathcal{L}_\star^{(2)*}$ in [Proposition 4.1](#).

To isolate the cutoff boundary, write the zero-extended microscopic Heisenberg generator as a sum over local supports $Z \subset \Lambda$. Let $\mathfrak{P}_Z^{\text{tr}}$ be the zero extension of the cutoff local superoperator supported on Z , and let $\mathfrak{P}_Z^{\text{full}}$ be the same polynomial superoperator evaluated with canonical full-Fock operators. Their moving-frame discrepancy is, by definition,

$$(4.9) \quad \mathcal{E}_{\Lambda,N}^{\text{bdry}}(t)A := W_N(t)^* \sum_Z (\mathfrak{P}_Z^{\text{tr}} - \mathfrak{P}_Z^{\text{full}})(W_N(t)AW_N(t)^*)W_N(t).$$

Below, $\mathfrak{P}^{\text{tr}} := \sum_Z \mathfrak{P}_Z^{\text{tr}}$ and $\mathfrak{P}^{\text{full}} := \sum_Z \mathfrak{P}_Z^{\text{full}}$. Within an estimate at a fixed time, the shorthand $\tilde{\rho}$ means $\tilde{\rho}_{\Lambda,N}(t)$.

Lemma 4.2. *Fix $\Lambda, T < \infty$, and a bounded coherent trajectory $u(t)$. For the projected coherent initial data (4.6), choose $c \geq c_T$ as in [Lemma 2.5](#). Let*

$$A_0 = \mathcal{P}(b, b^*), \quad G_t = W_N(t)A_0W_N(t)^*$$

for a fixed-degree fluctuation polynomial $\mathcal{P}$. Then there are $C_{A_0,\Lambda,T} < \infty$ and $c_{A_0} > 0$, independent of N and of all larger cutoff ratios c , such that

$$\sup_{0 \leq t \leq T} \left| \tilde{\rho}_{\Lambda,N}(t) \left(\mathcal{E}_{\Lambda,N}^{\text{bdry}}(t)A_0 \right) \right| \leq C_{A_0,\Lambda,T} e^{-c_{A_0}N}.$$

Proof. By the definition of the moving frame,

$$(4.10) \quad \tilde{\rho}_{\Lambda,N}(t) \left(\mathcal{E}_{\Lambda,N}^{\text{bdry}}(t) A_0 \right) = \rho_{\Lambda,N}(t) \left((\mathfrak{P}^{\text{tr}} - \mathfrak{P}^{\text{full}})(G_t) \right).$$

Here $\mathfrak{P}^{\text{tr}}$ is the fixed-degree local Heisenberg superoperator built from projected creation and annihilation operators, and $\mathfrak{P}^{\text{full}}$ is its zero-extended full-Fock counterpart. Let $\mathbf{P}_{\Lambda,M_N}$ denote the zero-extension projection onto the cutoff-supported tensor product. For some $m_{A_0} \in \mathbb{N}$ and a fixed buffer C_{buf} depending only on the polynomial degree and interaction range, telescoping each operator word gives the cutoff-supported identity

$$(4.11) \quad \mathbf{P}_{\Lambda,M_N} \left[(\mathfrak{P}^{\text{tr}} - \mathfrak{P}^{\text{full}})(G_t) \right] \mathbf{P}_{\Lambda,M_N} = \sum_{r=1}^{m_{A_0}} A_r(t) \Pi_{x_r,N}^{\text{bdry}} B_r(t), \quad \Pi_{x,N}^{\text{bdry}} = \mathbf{1}_{\{n_x \geq M_N - C_{\text{buf}}\}}.$$

Both the input and output in (4.11) lie in the cutoff-supported subspace. A word of degree d can cross the cutoff only within d occupation levels of the top state, and $X_1 \cdots X_m - Y_1 \cdots Y_m = \sum_j X_1 \cdots X_{j-1} (X_j - Y_j) Y_{j+1} \cdots Y_m$ places one cutoff defect in every summand. The commutator defect $[a_{M_N}, a_{M_N}^*] - I = -(M_N + 1)|M_N\rangle\langle M_N|$ is the simplest example.

Dissipative terms have the same support structure. Writing $L_M = L + \delta L$ after zero extension,

$$\begin{aligned} L_M^* G_t L_M - L^* G_t L &= \delta L^* G_t L_M + L^* G_t \delta L, \\ L_M^* L_M - L^* L &= \delta L^* L_M + L^* \delta L, \end{aligned}$$

and every δL is supported on the original-space boundary projector. Hamiltonian commutators, jump cross terms, and anticommutators therefore have the form (4.11). Each $A_r(t)$ and $B_r(t)$ is a finite sum of cutoff-restricted words of fixed degree. The degree bound depends only on the degree of A_0 and the local generator, while the coefficients are polynomials in the bounded trajectory amplitudes and in $\sqrt{N}$.

Weyl conjugation preserves polynomial degree but does not leave the coefficients N -independent: each displacement contributes a factor $\sqrt{N} \alpha_x(t)$. If d_{A_0} is the field degree of A_0 , boundedness of the trajectory and

$$\|a_x \mathbf{P}_{\Lambda,M_N}\|, \|a_x^* \mathbf{P}_{\Lambda,M_N}\| \leq \sqrt{M_N + 1}$$

give

$$(4.12) \quad \|\mathbf{P}_{\Lambda,M_N} G_t \mathbf{P}_{\Lambda,M_N}\| \leq C_{A_0,T} (\sqrt{N} + \sqrt{M_N + 1})^{d_{A_0}} \leq C_{A_0,T} (cN)^{d_{A_0}/2}.$$

After the fixed-degree factors $A_r(t)$ and $B_r(t)$ in (4.11) are included, Cauchy–Schwarz produces at most a polynomial factor $C_{A_0,\Lambda,T}(cN)^{p_{A_0}}$ for some finite p_{A_0} . Using (2.18), we obtain

$$C_{A_0,\Lambda,T}(cN)^{p_{A_0}} \rho_{\Lambda,N}(t) (\Pi_{x,N}^{\text{bdry}})^{1/2} \leq C_{A_0,\Lambda,T}(cN)^{p_{A_0}} \exp \left\{ -\frac{N}{2} [\theta c - m_T(\theta, \zeta)] \right\}.$$

Put $g(c) = \theta c - m_T(\theta, \zeta)$ and $\delta = g(c_T) > 0$. For $c \geq c_T$, $cN \leq C_{\theta,\delta} N g(c)$, so

$$(cN)^{p_{A_0}} e^{-Ng(c)/2} \leq C(Ng(c))^{p_{A_0}} e^{-Ng(c)/4} e^{-\delta N/4} \leq C' e^{-\delta N/4}.$$

Summation over the fixed torus proves the estimate uniformly over all larger cutoff ratios. $\square$

Lemma 4.3. *On every fixed torus and bounded coherent trajectory, define*

$$L_x^{(1)}(t) = 2\sqrt{\gamma} \alpha_x(t) b_x, \quad L_x^{(2)} = \sqrt{\gamma} b_x^2,$$

and, for two jump operators L, Q , set

$$(4.13) \quad \mathcal{B}_{L,Q}^*(A) = L^* A Q + Q^* A L - \frac{1}{2} \{L^* Q + Q^* L, A\}.$$

Then the shifted generator on fixed-degree fluctuation polynomials has the exact decomposition

$$(4.14) \quad \widehat{\mathcal{L}}_{\Lambda,N}^{\text{fl}}(t) = \mathcal{L}_{u(t)}^{(2)} + N^{-1/2} \sum_{x \in \Lambda} \mathcal{B}_{L_x^{(1)}(t), L_x^{(2)}}^* + \frac{\gamma}{N} \sum_{x \in \Lambda} \mathcal{D}[b_x^2]^* + \mathcal{E}_{\Lambda,N}^{\text{bdry}}(t).$$

The cross superoperator maps a quadratic fluctuation observable to a fluctuation polynomial of degree at most three, while the residual dissipator maps it to one of degree at most four. Their coefficients are uniform in N on $[0, T]$. For $G = (1 + \mathfrak{R}_b)^2$,

$$(4.15) \quad \left| \tilde{\rho} \left(\sum_x \mathcal{B}_{L_x^{(1)}, L_x^{(2)}}^* G \right) \right| \leq C \tilde{\rho}((1 + \mathfrak{R}_b)^{5/2}),$$

and the quartic term has the exact sign

$$(4.16) \quad \gamma \sum_x \mathcal{D}[b_x^2]^* G = -4\gamma \mathfrak{R}_b \sum_x b_x^{*2} b_x^2.$$

There is no additional quartic remainder away from the cutoff boundary.

Proof. Substitute $a_x = \sqrt{N}\alpha_x(t) + b_x$ in every local Hamiltonian and jump, and include the derivative term in (4.7). All Hamiltonians are at most quadratic and every jump other than the two-photon jump is linear, so after the coherent equation cancels the $O(\sqrt{N})$ terms they contribute only to the quadratic generator $\mathcal{L}_{u(t)}^{(2)}$ and to the cutoff-boundary error.

For the two-photon jump,

$$\sqrt{\frac{\gamma}{N}} a_x^2 = \sqrt{\gamma N} \alpha_x^2 + L_x^{(1)} + N^{-1/2} L_x^{(2)}.$$

The scalar part is removed by the standard Lindblad gauge and combines with the moving-frame Hamiltonian. Expanding the remaining dissipator gives the exact identity

$$\mathcal{D}[L_x^{(1)} + N^{-1/2} L_x^{(2)}]^* = \mathcal{D}[L_x^{(1)}]^* + N^{-1/2} \mathcal{B}_{L_x^{(1)}, L_x^{(2)}}^* + N^{-1} \mathcal{D}[L_x^{(2)}]^*.$$

Since $\mathcal{D}[L_x^{(2)}]^* = \gamma \mathcal{D}[b_x^2]^*$, summing over x proves (4.14).

For every field monomial M of degree r on the fixed torus,

$$(4.17) \quad \|M(1 + \mathfrak{R}_b)^{-r/2}\| \leq C_{r, \Lambda}.$$

Normal ordering the cubic cross term applied to G , followed by (4.17) and Cauchy–Schwarz, proves (4.15). Finally b_x^2 lowers $\mathfrak{R}_b$ by two while $b_x^{*2} b_x^2$ commutes with it, so

$$\mathcal{D}[b_x^2]^* G = b_x^{*2} b_x^2 [(1 + \mathfrak{R}_b - 2)^2 - (1 + \mathfrak{R}_b)^2] = -4\mathfrak{R}_b b_x^{*2} b_x^2,$$

which proves (4.16). $\square$

Lemma 4.4. Fix a finite torus Λ , a bounded coherent trajectory $u(t)$ on $0 \leq t \leq T$, and the projected coherent initial data (4.6). Choose $c \geq c_T$ as above. Then

$$\sup_{N \geq 1} \sup_{0 \leq t \leq T} \tilde{\rho}_{\Lambda, N}(t)((1 + \mathfrak{R}_b)^2) \leq C_{\Lambda, T, u},$$

uniformly over every cutoff family with ratio $c \geq c_T$.

Proof. Apply Lemma 4.3 to $G = (1 + \mathfrak{R}_b)^2$. On the fixed torus write

$$\mathcal{L}_{u(t)}^{(2)} = i[H^{(2)}(t), \cdot] + \sum_{\kappa=1}^{M_\Lambda} \mathcal{D}[L_\kappa^{(1)}(t)]^*,$$

where $M_\Lambda \in \mathbb{N}$ is the finite number of linear jumps in this chosen representation, $H^{(2)}(t)$ is quadratic, every $L_\kappa^{(1)}(t)$ is linear in the fluctuation fields, and all coefficients are uniformly bounded on $[0, T]$. Normal ordering $i[H^{(2)}, G]$ and each $\mathcal{D}[L_\kappa^{(1)}]^* G$ produces only finitely many monomials of degree at most four. By (4.17), Cauchy–Schwarz, and the form inequality

$$|\tilde{\rho}(M + M^*)| \leq C_M \tilde{\rho}((1 + \mathfrak{R}_b)^2) + C_M \quad (\deg M \leq 4),$$

we obtain

$$(4.18) \quad \left| \tilde{\rho} \left(\mathcal{L}_{u(t)}^{(2)} (1 + \mathfrak{N}_b)^2 \right) \right| \leq C \tilde{\rho}((1 + \mathfrak{N}_b)^2) + C.$$

The boundary discrepancy is exponentially small by [Lemma 4.2](#). Writing $\mathfrak{n}_x = b_x^* b_x$, the commuting number operators satisfy on the finite-particle core

$$\begin{aligned} \sum_x b_x^{*2} b_x^2 &= \sum_x \mathfrak{n}_x (\mathfrak{n}_x - 1) \geq |\Lambda|^{-1} \mathfrak{N}_b^2 - \mathfrak{N}_b, \\ \mathfrak{N}_b \sum_x b_x^{*2} b_x^2 &\geq |\Lambda|^{-1} \mathfrak{N}_b^3 - \mathfrak{N}_b^2. \end{aligned}$$

The residual two-photon drift controls $N^{-1}(1 + \mathfrak{N}_b)^3$ modulo $C(1 + \mathfrak{N}_b)^2$. For every $\epsilon_Y > 0$, the scalar Young inequality

$$N^{-1/2}(1 + s)^{5/2} \leq \epsilon_Y N^{-1}(1 + s)^3 + C_{\epsilon_Y}(1 + s)^2$$

absorbs the cubic remainder into this negative drift. By [Lemma 4.3](#), this is the only quartic term: the remaining nonlinear contribution is the $N^{-1/2}$ cubic cross term, which is absorbed by the preceding Young inequality. Consequently

$$\tilde{\rho}_{\Lambda, N}(t)(\tilde{\mathcal{L}}_{\Lambda, N}^{\text{fl}}(t)G) \leq C \tilde{\rho}_{\Lambda, N}(t)(G) + C + C e^{-c_{\text{tail}} N}.$$

The unprojected coherent product is the moving-frame vacuum at $t = 0$, and projection changes fixed fluctuation moments only by an exponentially small original-occupation tail. Gronwall proves the bound. $\square$

Let Λ be fixed, let $u(t)$ be a bounded first-moment solution on $0 \leq t \leq T$, and choose a cutoff ratio $c \geq c_T$ supplied by [Lemma 2.5](#). Start the microscopic evolution from (4.6). Define

$$\Xi_N(t) = (\sqrt{N}(Q_{x, N} - q_x(t)), \sqrt{N}(P_{x, N} - p_x(t)))_{x \in \Lambda}^\top$$

and

$$(V_N(t))_{ij} = \frac{1}{2} \rho_{\Lambda, N}(t) \left(\bar{\Xi}_{N, i}(t) \bar{\Xi}_{N, j}(t) + \bar{\Xi}_{N, j}(t) \bar{\Xi}_{N, i}(t) \right), \quad \bar{\Xi}_N(t) = \Xi_N(t) - \rho_{\Lambda, N}(t)(\Xi_N(t)).$$

Set $A(t) := D_u \mathcal{F}_\lambda^\Lambda(u(t))$. If $L_j^\Lambda = -\Delta_j^\Lambda$ denotes the directional graph Laplacian on the fixed torus, define

$$D(t) = \bigoplus_{x \in \Lambda} \left[\frac{\kappa}{2} + \gamma(q_x(t)^2 + p_x(t)^2) \right] I_2 + \sum_{j=1}^d K_j L_j^\Lambda \otimes I_2.$$

Let $V(t)$ solve

$$(4.19) \quad \dot{V}(t) = A(t)V(t) + V(t)A(t)^\top + D(t), \quad V(0) = \frac{1}{2} I_{2|\Lambda|}.$$

Theorem 4.5 (Covariance convergence). *Under the preceding assumptions, for every fixed matrix norm there is a constant $C_{\Lambda, T, u}$ such that, for all $c \geq c_T$,*

$$\sup_{0 \leq t \leq T} \|V_N(t) - V(t)\| \leq C_{\Lambda, T, u} N^{-1/2} + C_{\Lambda, T, u} e^{-c_{\text{tail}} N}.$$

The constant is independent of c and N .

Proof. Apply (4.14) to every centered symmetrized quadratic monomial. The moving-frame derivative cancels all $O(\sqrt{N})$ linear terms by the coherent equation. Componentwise one obtains

$$(4.20) \quad \frac{d}{dt} (V_N)_{ij} = (A(t)V_N + V_N A(t)^\top + D(t))_{ij} + N^{-1/2} \varepsilon_{ij}^{(3)}(t) + N^{-1} \varepsilon_{ij}^{(4)}(t) + \varepsilon_{ij}^{\text{bdry}}(t).$$

Here $\varepsilon^{(3)}$ is a finite linear combination of centered fluctuation moments of degree at most three, and $\varepsilon^{(4)}$ one of degree at most four. A representative component is the local quadratic observable $\mathbf{n}_x = b_x^* b_x$. From (4.13),

$$(4.21) \quad \mathcal{B}_{L_x^{(1)}, L_x^{(2)}}^*(\mathbf{n}_x) = -3\gamma(\bar{\alpha}_x b_x^* b_x^2 + \alpha_x b_x^{*2} b_x), \quad \gamma \mathcal{D}[b_x^2]^*(\mathbf{n}_x) = -2\gamma b_x^{*2} b_x^2.$$

Its uncentered second-moment equation contains these $N^{-1/2}$ cubic and N^{-1} quartic terms. Passing to the centered covariance adds products of the corresponding linear remainders; the first-moment estimate below bounds them with those orders. Every other symmetrized quadratic monomial is one of finitely many analogous commutator calculations. The normal-ordering inequality (4.17) and Lemma 4.4 imply

$$(4.22) \quad \sup_{0 \leq t \leq T} (|\varepsilon_{ij}^{(3)}(t)| + |\varepsilon_{ij}^{(4)}(t)|) \leq C_{\Lambda, T, u}, \quad |\varepsilon_{ij}^{\text{bdry}}(t)| \leq C_{\Lambda, T, u} e^{-c_{\text{tail}} N}.$$

The diffusion term in (4.20) is obtained directly from the linearized one-photon, two-photon, and bond jumps and is the matrix $D(t)$ stated in the theorem.

Applying the displaced-generator expansion to the linear fluctuation vector yields

$$\dot{m}_N(t) = A(t)m_N(t) + r_N^{(1)}(t), \quad m_N(t) = \rho_{\Lambda, N}(t)(\Xi_N(t)),$$

with $\sup_t \|r_N^{(1)}(t)\| \leq CN^{-1/2} + Ce^{-c_{\text{tail}} N}$. Projected coherent initial data give $m_N(0) = O(e^{-c_{\text{tail}} N})$, so variation of constants yields $\sup_t \|m_N(t)\| \leq CN^{-1/2} + Ce^{-c_{\text{tail}} N}$. Subtracting $m_N m_N^\top$ from the uncentered second moment changes (4.20) only by a remainder at these orders.

In matrix form,

$$\dot{V}_N = A(t)V_N + V_N A(t)^\top + D(t) + R_N(t), \quad \sup_{0 \leq t \leq T} \|R_N(t)\| \leq C_{\Lambda, T, u} N^{-1/2} + C_{\Lambda, T, u} e^{-c_{\text{tail}} N}.$$

The projected coherent initial covariance differs from $I/2$ by an exponentially small cutoff tail. Since $A(t)$ is bounded on the fixed interval, its propagator is uniformly bounded for $0 \leq s \leq t \leq T$. Variation of constants proves the asserted covariance estimate. $\square$

For n_{modes} bosonic modes, write

$$(4.23) \quad J_{n_{\text{modes}}} := \bigoplus_{j=1}^{n_{\text{modes}}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a real two-mode covariance $W \in \mathbb{R}^{4 \times 4}$, partial transpose on the second mode is represented by

$$(4.24) \quad \Lambda_{\text{PT}} := \text{diag}(1, 1, 1, -1), \quad W^\Gamma := \Lambda_{\text{PT}} W \Lambda_{\text{PT}}, \quad H_{\text{PT}}(W) := W^\Gamma + \frac{i}{2} J_2.$$

The partially transposed symplectic eigenvalues are the positive moduli of the eigenvalues of $iJ_2 W^\Gamma$. Define the smaller one and the strict uncertainty-violation margin by

$$(4.25) \quad \tilde{\nu}_-(W) := \min\{|\lambda| : \lambda \in \text{Spec}(iJ_2 W^\Gamma)\}, \quad \delta_{\text{PT}}(W) := -\lambda_{\min}(H_{\text{PT}}(W)).$$

The partially transposed covariance violates the Robertson–Schrödinger uncertainty relation exactly when $\tilde{\nu}_-(W) < 1/2$, equivalently $\delta_{\text{PT}}(W) > 0$. For a two-mode Gaussian state this is equivalent to NPT; for an arbitrary state the violation is a sufficient NPT witness.

A matrix $\mathbf{T} \in \mathbb{R}^{4 \times 2n_{\text{modes}}}$ is called a passive canonical two-mode selector when

$$(4.26) \quad \mathbf{T} \mathbf{T}^\top = I_4, \quad \mathbf{T} J_{n_{\text{modes}}} \mathbf{T}^\top = J_2,$$

and there exists an orthogonal symplectic matrix $\mathbf{S} \in O(2n_{\text{modes}}) \cap \text{Sp}(2n_{\text{modes}}, \mathbb{R})$ whose first four rows are $\mathbf{T}$. This extension condition identifies the selector with a passive mode unitary followed by tracing out the unselected modes.

Corollary 4.6. *In the setting of Theorem 4.5, first choose any cutoff ratio $c \geq c_T$. Let $t_* \in (0, T]$ and let $\mathsf{T} \in \mathbb{R}^{4 \times 2|\Lambda|}$ be a passive canonical two-mode selector in the preceding sense. If*

$$\tilde{v}_-(\mathsf{TV}(t_*)\mathsf{T}^\mathsf{T}) < \frac{1}{2},$$

then, for the chosen cutoff ratio, there exists $N_0 < \infty$ such that every $N \geq N_0$ has an NPT selected two-mode reduction at time t_ .*

Proof. Zero-extend the locally truncated state, apply a passive mode unitary implementing a symplectic extension of T , and trace out the unselected modes. Choose $0 < \delta < \delta_{\mathsf{PT}}(\mathsf{TV}(t_*)\mathsf{T}^\mathsf{T})$. After the cutoff ratio has been fixed, the covariance convergence theorem transfers that violation to the microscopic covariance for all sufficiently large N . Every PPT two-mode state obeys the Robertson–Schrödinger uncertainty relation after partial transpose, so the selected two-mode reduced state is NPT. $\square$

Corollary 4.7. *Let $\Lambda = C$, let $u(t) = u^*$ be a stationary periodic branch, and assume that A_{per} is Hurwitz. Write $V_{N,c}(t)$ for the microscopic covariance with cutoff $M_N = \lfloor cN \rfloor$. For every tolerance $\epsilon_{\text{tol}} > 0$, there are $t_{\epsilon_{\text{tol}}} < \infty$ and $c_{\epsilon_{\text{tol}}} < \infty$ such that, for every $c \geq c_{\epsilon_{\text{tol}}}$,*

$$\limsup_{N \rightarrow \infty} \|V_{N,c}(t_{\epsilon_{\text{tol}}}) - V_{\text{per}}\| \leq \epsilon_{\text{tol}}.$$

If a passive canonical two-mode selector T satisfies

$$\tilde{v}_-(\mathsf{TV}_{\text{per}}\mathsf{T}^\mathsf{T}) < \frac{1}{2},$$

then there are finite t_0, c_0, N_0 such that, for every $c \geq c_0$ and $N \geq N_0$, the selected two-mode reduced state of the canonically embedded microscopic state at time t_0 is NPT.

Proof. For a stationary branch, (4.19) reduces to (4.5), whose solution converges exponentially to V_{per} . Given $\epsilon_{\text{tol}} > 0$, first choose $t_{\epsilon_{\text{tol}}}$ so that $\|V(t_{\epsilon_{\text{tol}}}) - V_{\text{per}}\| < \epsilon_{\text{tol}}/2$. Next choose $c_{\epsilon_{\text{tol}}} \geq c_{t_{\epsilon_{\text{tol}}}}$ from Theorem 4.5. The covariance-convergence estimate and the triangle inequality give the displayed bound. If $\mathsf{TV}_{\text{per}}\mathsf{T}^\mathsf{T}$ has a strict NPT margin, continuity of the partially transposed symplectic spectrum gives a sufficiently late finite t_0 for the limiting Gaussian covariance; choosing the finite relaxation time t_0 , then $c_0 \geq c_{t_0}$, and finally N_0 allows Corollary 4.6 to transfer that strict violation to the microscopic two-mode reduction. $\square$

For a one-dimensional period cell, set $C^* := (2\pi/|C|)(\mathbb{Z}/|C|\mathbb{Z})$. Choose $k \in C^*$ with $k \neq 0$ and $k \not\equiv -k \pmod{2\pi}$, thereby excluding the self-conjugate modes 0 and, when present, π . Define $b_k := |C|^{-1/2} \sum_{x \in C} e^{-ikx} b_x$. The corresponding real quadratures are

$$(4.27) \quad \begin{pmatrix} Q_k \\ P_k \\ Q_{-k} \\ P_{-k} \end{pmatrix} = \frac{1}{\sqrt{|C|}} \sum_{x \in C} \begin{pmatrix} \cos(kx) & \sin(kx) \\ -\sin(kx) & \cos(kx) \\ \cos(kx) & -\sin(kx) \\ \sin(kx) & \cos(kx) \end{pmatrix} \begin{pmatrix} \delta q_x \\ \delta p_x \end{pmatrix} =: \mathsf{T}_{k,-k} \xi.$$

The Fourier rows obey $\mathsf{T}_{k,-k} \mathsf{T}_{k,-k}^\mathsf{T} = I_4$ and $\mathsf{T}_{k,-k} J_{|C|} \mathsf{T}_{k,-k}^\mathsf{T} = J_2$; completing the real Fourier transform gives the required orthogonal symplectic extension. Thus $\mathsf{T}_{k,-k}$ is a passive canonical selector, and

$$V_{k,-k} := \mathsf{T}_{k,-k} V_{\text{per}} \mathsf{T}_{k,-k}^\mathsf{T}, \quad \tilde{v}_-(k, -k) := \tilde{v}_-(V_{k,-k}).$$

By the two-mode Gaussian PPT criterion [30–33], the pair is NPT exactly when $\tilde{v}_-(k, -k) < 1/2$, with logarithmic negativity

$$E_{\text{LN}}(k, -k) := \max\{0, -\log_2[2\tilde{v}_-(k, -k)]\}.$$

Define the standing-wave quadratures by

$$Q_+ := \frac{Q_k + Q_{-k}}{\sqrt{2}}, \quad P_+ := \frac{P_k + P_{-k}}{\sqrt{2}}, \quad Q_- := \frac{Q_k - Q_{-k}}{\sqrt{2}}, \quad P_- := \frac{P_k - P_{-k}}{\sqrt{2}}.$$

In the standing-wave order (Q_+, P_+, Q_-, P_-) associated with $A_k^{(+)} \oplus A_k^{(-)}$ and the traveling-mode order $(Q_k, P_k, Q_{-k}, P_{-k})$, the passive transformation is

$$(4.28) \quad S_{\text{sw} \rightarrow \text{tr}} := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \quad S_{\text{sw} \rightarrow \text{tr}} \in O(4) \cap \text{Sp}(4, \mathbb{R}).$$

Proposition 4.8. *At the homogeneous state $u^\star = 0$, set*

$$\omega_j(k) = 2 - 2 \cos k_j, \quad \kappa_k = \kappa + 2 \sum_{j=1}^d K_j \omega_j(k), \quad g_k = \varepsilon - \sum_{j=1}^d K'_j \omega_j(k),$$

$$R_k = \sqrt{\kappa_k^2 + 4\Omega_\lambda^2}, \quad \eta_k = \frac{2|g_k|}{R_k}, \quad \Delta_k = R_k^2 - 4g_k^2, \quad s_k = \frac{R_k^2}{2\Delta_k}.$$

For a non-self-conjugate pair $(k, -k)$, a real standing-wave basis gives

$$(4.29) \quad A_{k,-k}^{\text{sw}} = A_k^{(+)} \oplus A_k^{(-)}, \quad D_{k,-k}^{\text{sw}} = \frac{\kappa_k}{2} I_4,$$

where

$$(4.30) \quad A_k^{(\pm)} = \begin{pmatrix} -\kappa_k/2 \pm g_k & \Omega_\lambda \\ -\Omega_\lambda & -\kappa_k/2 \mp g_k \end{pmatrix}, \quad D_k^{(\pm)} = \frac{\kappa_k}{2} I_2.$$

Here $A_k^{(\pm)}$ is the classical Fourier symbol in the chosen real quadratures. If $\eta_k < 1$, the stationary covariance of the traveling modes (b_k, b_{-k}) , in the order $(Q_k, P_k, Q_{-k}, P_{-k})$, is

$$(4.31) \quad V_{k,-k}^{\text{hom}} = \begin{pmatrix} s_k I_2 & C_k \\ C_k & s_k I_2 \end{pmatrix}, \quad C_k = \frac{g_k}{\Delta_k} \begin{pmatrix} \kappa_k & -2\Omega_\lambda \\ -2\Omega_\lambda & -\kappa_k \end{pmatrix}.$$

Both ordinary symplectic eigenvalues are

$$(4.32) \quad \nu_{\text{phys}}(k) = \frac{R_k}{2\sqrt{\Delta_k}} = \frac{1}{2\sqrt{1-\eta_k^2}}.$$

Theorem 4.9 (Homogeneous stability and opposite-momentum entanglement). *Let $(k, -k)$ be a non-self-conjugate homogeneous momentum pair and use the quantities in Proposition 4.8. The pair is stable if and only if $\eta_k < 1$. In the stable regime,*

$$(4.33) \quad \tilde{\nu}_-(k, -k) = \frac{R_k}{2(R_k + 2|g_k|)}, \quad \tilde{\nu}_+(k, -k) = \frac{R_k}{2(R_k - 2|g_k|)},$$

and hence

$$(4.34) \quad \tilde{\nu}_-(k, -k) = \frac{1}{2(1 + \eta_k)}, \quad E_{\text{LN}}(k, -k) = \log_2(1 + \eta_k).$$

The classical Fourier block satisfies

$$\det A_k^{(+)} = \frac{R_k^2}{4}(1 - \eta_k^2).$$

Proof of Proposition 4.8 and Theorem 4.9. The homogeneous Fourier transform decomposes the pair $(k, -k)$ into two orthogonal standing-wave modes with equal damping κ_k and opposite single-mode squeezing parameters $\pm g_k$, as in (4.29)–(4.30). Solving the two 2×2 Lyapunov equations yields

$$V_k^{(+)} = \frac{1}{\Delta_k} \begin{pmatrix} R_k^2/2 + \kappa_k g_k & -2\Omega_\lambda g_k \\ -2\Omega_\lambda g_k & R_k^2/2 - \kappa_k g_k \end{pmatrix}, \quad V_k^{(-)} = V_k^{(+)}|_{g_k \mapsto -g_k}.$$

Both drift blocks have trace $-\kappa_k < 0$ and determinant $\Delta_k/4$, so their Hurwitz condition is $\eta_k < 1$. The explicit passive symplectic matrix $S_{\text{sw} \rightarrow \text{tr}}$ in (4.28) recombines the two standing waves into the traveling

modes (b_k, b_{-k}) and sends $V_k^{(+)} \oplus V_k^{(-)}$ to (4.31). A direct symplectic calculation then gives the repeated ordinary eigenvalue (4.32). Partial transpose changes the sign of P_{-k} and gives (4.33); the smaller eigenvalue and the logarithmic negativity are then (4.34). $\square$

Corollary 4.10. *Suppose every linear damping channel contributing to a stable homogeneous pair is coupled to a common thermal occupation $\bar{n} \geq 0$. The drift is unchanged and the diffusion is multiplied by $2\bar{n} + 1$. Then*

$$(4.35) \quad \tilde{\nu}_{-}^{(\bar{n})}(k, -k) = \frac{2\bar{n} + 1}{2(1 + \eta_k)}, \quad E_{\text{LN}}^{(\bar{n})}(k, -k) = \max \left\{ 0, \log_2 \frac{1 + \eta_k}{2\bar{n} + 1} \right\}.$$

In particular

$$\text{NPT} \iff \eta_k > 2\bar{n}.$$

Proof. The Lyapunov equation is linear in the diffusion matrix, so the stationary covariance in (4.31) is multiplied by $2\bar{n} + 1$. Symplectic eigenvalues scale by this factor, and (4.35) follows from (4.34). $\square$

The bipartition in Theorem 4.9 is the physically distinguished pair of counterpropagating Fourier modes b_k and b_{-k} . The parametric term squeezes the two standing-wave modes in orthogonal quadratures; their passive recombination into traveling modes therefore produces two-mode NPT entanglement whenever $g_k \neq 0$.

Corollary 4.11. *In the homogeneous Gaussian sector, every stable non-self-conjugate pair with $g_k \neq 0$ is NPT. Within the stable homogeneous phase, minimizing $\tilde{\nu}_{-}(k, -k)$, maximizing $E_{\text{LN}}(k, -k)$, and maximizing η_k are equivalent spectral optimizations. The classical determinant is*

$$(4.36) \quad \det A_k^{(+)} = \frac{R_k^2}{4}(1 - \eta_k^2),$$

so the Turing threshold at k_* is the condition $\eta_{k_*} = 1$. Consequently

$$\lambda \uparrow 0 : \quad \nu_{\text{phys}}(k_*) \longrightarrow \infty, \quad \tilde{\nu}_{-}(k_*, -k_*) \longrightarrow \frac{1}{4}, \quad E_{\text{LN}}(k_*, -k_*) \longrightarrow 1.$$

No finite stationary covariance exists at threshold because the critical drift mode is neutral, but the two NPT quantities have the displayed one-sided limits.

Proof. Equations (4.32)–(4.34) are monotone functions of η_k , while (4.36) follows from the determinant in (4.30). $\square$

Corollary 4.12. *If the transport is scalar, then $K'_j = 0$ for every j and $g_k = \varepsilon$ is momentum independent. Hence η_k decreases with the weighted transport symbol*

$$\mathbf{t}(k) := \sum_{j=1}^d K_j \omega_j(k),$$

and the strongest opposite-momentum NPT occurs at nonzero modes minimizing $\mathbf{t}(k)$. In the isotropic scalar case these are the smallest nonzero momenta. For the explicit differential-transport stripe family on the stable side, the longitudinal maximum of η_k occurs at

$$(4.37) \quad \omega_{\text{ent}}(\lambda) = 2 - \sqrt{3} - \left(\frac{2}{\sqrt{3}} - 1 \right) \lambda, \quad \lambda < 0,$$

for λ sufficiently close to zero. The homogeneous NPT minimum is at a finite wave number and converges to the Turing value $\omega_* = 2 - \sqrt{3}$ as $\lambda \uparrow 0$.

Proof. Under scalar transport, R_k increases monotonically with $\mathbf{t}(k)$ while $|g_k|$ is fixed. For the explicit stripe parameters, differentiation of η_k^2 with respect to the longitudinal lattice symbol ω gives a unique interior maximum at (4.37). $\square$

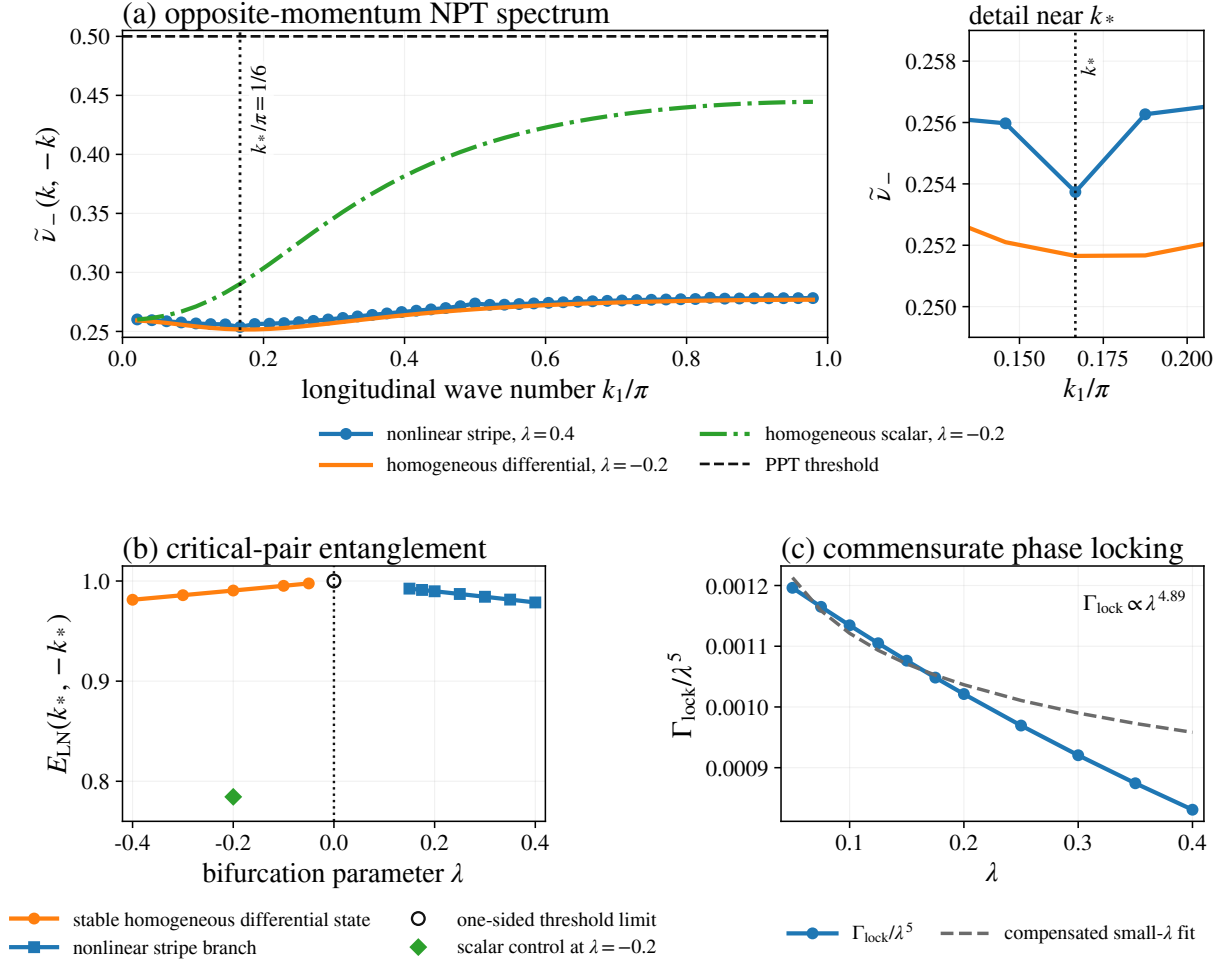


FIGURE 4. Momentum-resolved Gaussian entanglement and phase locking. (a) Longitudinal opposite-momentum spectra for the nonlinear stripe, the homogeneous differential-transport system, and a scalar-transport control; the detail panel resolves the neighborhood of k_* . (b) Critical-pair logarithmic negativity across the homogeneous threshold and along the continued patterned branch. (c) Phase-locking rate after division by the symmetry-predicted λ^5 scale.

Starting from the bifurcating branch, Newton continuation reaches the representative nonlinear value $(\lambda, \nu) = (0.4, 4)$. We apply [Proposition 4.1](#) to the unseeded generator ($h = 0$) and the unseeded bond-centered stationary stripe. At this value the transverse-uniform drift is Hurwitz with

$$\max\{\text{Re } z : z \in \text{Spec}(A_{\text{per}})\} = -8.51 \times 10^{-6}.$$

Let V_{per} denote the exact solution of $A_{\text{per}}V + VA_{\text{per}}^T + D_{\text{per}} = 0$, and let $\widehat{V}_{\text{per}}$ denote the floating-point Lyapunov solution used in the computation. For a real $n \times n$ matrix A , define the Lyapunov operator and its two-norm separation by

$$(4.38) \quad \mathcal{K}_A(X) := AX + XA^T, \quad \text{sep}_2(\mathcal{K}_A) := \sigma_{\min}(I_n \otimes A + A \otimes I_n),$$

where $\sigma_{\min}$ denotes the smallest singular value. The floating-point solution has relative residual

$$r_{\text{Lyap}} := \frac{\|A_{\text{per}}\widehat{V}_{\text{per}} + \widehat{V}_{\text{per}}A_{\text{per}}^T + D_{\text{per}}\|_F}{\|D_{\text{per}}\|_F} = 7.36 \times 10^{-11}.$$

The vectorized Lyapunov operator has two-norm separation $\text{sep}_2(\mathcal{K}_{A_{\text{per}}}) = 9.92 \times 10^{-6}$. The absolute Frobenius residual 3.64×10^{-9} and the separation estimate yield the a posteriori error bound

$$(4.39) \quad \|V_{\text{per}} - \widehat{V}_{\text{per}}\|_F \leq 3.66 \times 10^{-4}.$$

The numerical covariance is physical, with minimum symplectic eigenvalue 0.823. The critical pair gives

$$\widetilde{v}_-(k_*, -k_*) = 0.254, \quad E_{\text{LN}}(k_*, -k_*) = 0.979.$$

Let

$$W_*^{\text{exact}} := T_{k_*, -k_*} V_{\text{per}} T_{k_*, -k_*}^\top, \quad \widehat{W}_* := T_{k_*, -k_*} \widehat{V}_{\text{per}} T_{k_*, -k_*}^\top,$$

and define $H_{\text{PT}}^{\text{exact}} := H_{\text{PT}}(W_*^{\text{exact}})$ and $\widehat{H}_{\text{PT}} := H_{\text{PT}}(\widehat{W}_*)$. Because the selector has orthonormal rows and Λ_{PT} is orthogonal,

$$\|W_*^{\text{exact}} - \widehat{W}_*\|_2 \leq \|V_{\text{per}} - \widehat{V}_{\text{per}}\|_2 \leq \|V_{\text{per}} - \widehat{V}_{\text{per}}\|_F, \quad \|H_{\text{PT}}^{\text{exact}} - \widehat{H}_{\text{PT}}\|_2 = \|W_*^{\text{exact}} - \widehat{W}_*\|_2.$$

The numerical value is $\lambda_{\min}(\widehat{H}_{\text{PT}}) = -0.246$ using unrounded data, so Weyl's inequality and (4.39) give

$$\lambda_{\min}(H_{\text{PT}}^{\text{exact}}) \leq -0.246 + 3.66 \times 10^{-4} < -0.245 < 0.$$

An independent high-precision PPT invariant in the reproducibility archive confirms the sign. [Corollary 4.7](#) transfers this strict violation to sufficiently large microscopic systems initialized on the projected coherent stripe.

For the explicit stripe at $\lambda = 0.4$, this pair attains the global minimum among the resolved longitudinal opposite-momentum pairs $k = (k_1, 0)$ with $0 < k_1 < \pi$ in the transverse-uniform $L = 48$ and $L = 96$ supercells. The minimum among nearest-neighbor real-space two-mode reductions is instead $\widetilde{v}_- = 0.870 > 1/2$.

The closed formulas in [Theorem 4.9](#) and [Corollaries 4.11](#) and [4.12](#) explain the two homogeneous controls at the chosen parameter value. At $\lambda = -0.2$, write $\omega_{\text{ent}} := \omega_{\text{ent}}(-0.2)$ and let $k_{\text{ent}} \in (0, \pi)$ be the unique longitudinal wave number determined by

$$2(1 - \cos k_{\text{ent}}) = \omega_{\text{ent}}.$$

The continuous differential-transport spectrum is minimized at

$$\omega_{\text{ent}} = 0.299, \quad k_{\text{ent}}/\pi = 0.176,$$

which lies close to $k_*/\pi = 1/6$; on the $L = 96$ longitudinal momentum grid the minimum occurs at k_* and equals $\widetilde{v}_- = 0.252$. Keeping Ω_λ fixed and using scalar transport $D_q = D_p = 2 + \sqrt{3}$, the minimum moves to the smallest nonzero longitudinal momentum, $k_1/\pi = 1/48$, where $\widetilde{v}_- = 0.260$; at k_* the value is 0.290. Differential transport therefore relocates the homogeneous opposite-momentum NPT minimum from the long-wavelength edge to a finite wave number, while the representative nonlinear stripe retains the commensurate minimum in the transverse-uniform supercells examined here.

The leading drift eigenvector has overlap 0.997 with the phase tangent $-r \sin(k_* x + \pi/12)$, identifying the soft spectral abscissa as the commensurate phase-locking rate. Its observed scaling is consistent with the first symmetry-allowed C_{12} locking term.

4.3. Finite-time fluctuation spectra for spot and labyrinth. To obtain the fluctuation spectra for the spot and labyrinth regimes, we propagate the nonautonomous Lyapunov equation along each first-moment trajectory $u(t)$:

$$\dot{V}(t) = A(t)V(t) + V(t)A(t)^\top + D(t), \quad A(t) := D_u \mathcal{F}_\lambda^{\Lambda_L}(u(t)), \quad V(0) = \frac{1}{2}I_{2L^2},$$

where $A(t), D(t), V(t) \in \mathbb{R}^{2L^2 \times 2L^2}$ and $D(t)$ is assembled from the one-photon, two-photon, and dissipative-bond jumps as in (4.5). Let $U(t, s) \in \mathbb{R}^{2L^2 \times 2L^2}$ be the fundamental matrix defined by

$$\partial_t U(t, s) = A(t)U(t, s), \quad U(s, s) = I_{2L^2}.$$

Only selected 4×4 Fourier-pair reductions are required. Let $\mathsf{T}_{k,-k} \in \mathbb{R}^{4 \times 2L^2}$ be the passive selector for $(k, -k)$ and define $Z_k(\tau) := U(T, T - \tau)^\top \mathsf{T}_{k,-k}^\top \in \mathbb{R}^{2L^2 \times 4}$. Variation of constants yields the low-rank identity

$$\mathsf{T}_{k,-k} V(T) \mathsf{T}_{k,-k}^\top = \frac{1}{2} Z_k(T)^\top Z_k(T) + \int_0^T Z_k(\tau)^\top D(T - \tau) Z_k(\tau) d\tau.$$

We integrate the adjoint equation by Strang splitting: the translation-invariant linear part is exponentiated analytically in Fourier space and the local nonlinear Jacobian is exponentiated pointwise. The noise integral uses the trapezoidal rule. This avoids constructing the full $2L^2 \times 2L^2$ covariance.

At the finest covariance step, all seven sampled covariances are physical before partial transpose, and each selected-shell direction is NPT. The shell remains separated from both radial controls under covariance-step refinement and coherent-step halving. The minimum shell/control gaps are 0.010 for Spot and 0.005 for Labyrinth; the largest changes caused by halving Δt_{PDE} are 2.44×10^{-4} and 1.25×10^{-3} , respectively. Direction-resolved values and Richardson extrapolations are included in the reproducibility archive. [Corollary 4.6](#) transfers each strict finite-time violation to sufficiently large microscopic systems.

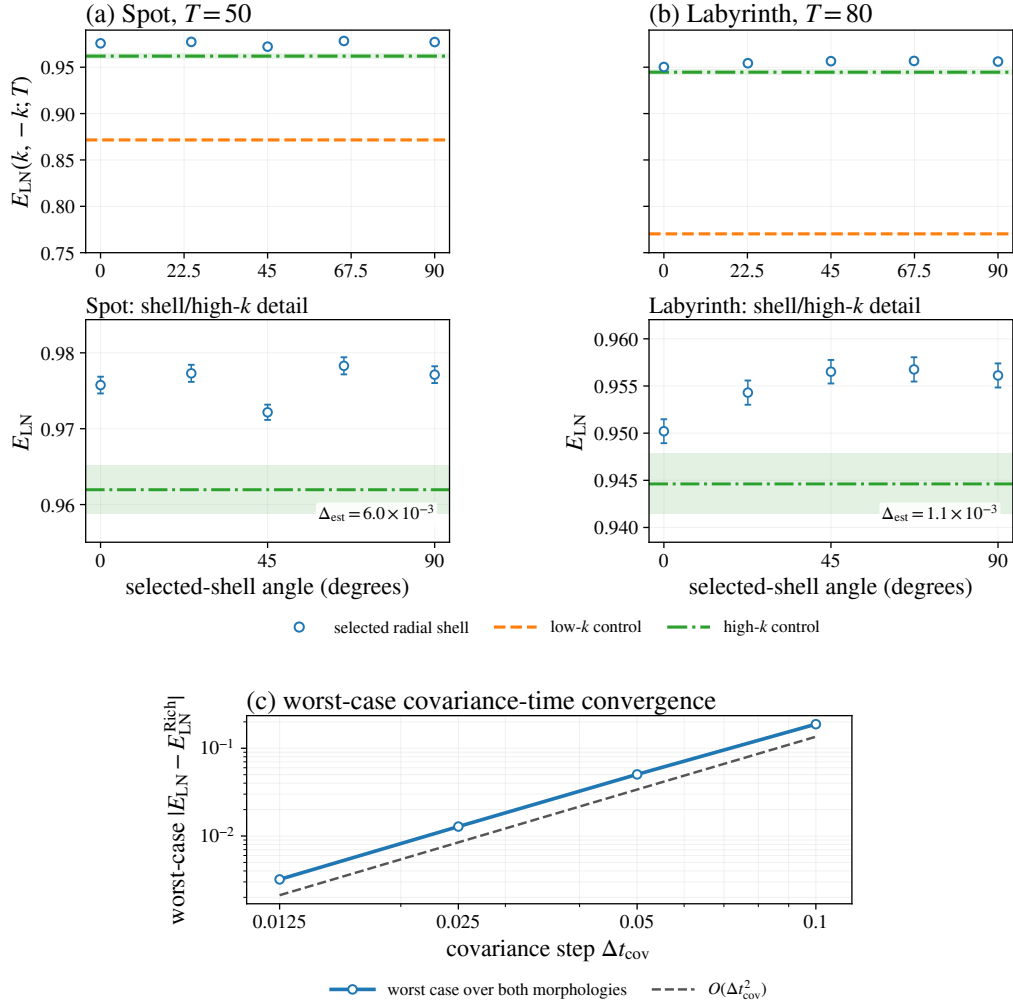


FIGURE 5. Opposite-momentum entanglement along the finite-time Spot and Labyrinth trajectories. Panels (a) and (b) compare five directions on the selected radial shell with low- and high-wave-number controls. Panel (c) shows the covariance-step differences for the sampled modes.

5. CONCLUSION

Completely positive lattice dynamics can select a nonzero Turing scale and sustain quantum order at the microscopic level. Our explicit family combines a supercritical bifurcation, stable commensurate stripe branches, and extensive Bragg order with a controlled fluctuation limit. Microscopic covariances converge at rate $O(N^{-1/2})$ to the Gaussian Lyapunov flow, carrying strict partial-transpose uncertainty violations to large N .

In the homogeneous Gaussian sector, a single spectral ratio governs both classical stability and opposite-momentum entanglement. The two-dimensional stripe, spot, and labyrinth computations show the same selected scale organizing morphology and quantum correlations. Together, the construction and fluctuation theory establish Lindblad dynamics as a microscopic origin of quantum Turing order.

ACKNOWLEDGMENTS

This work was supported in part by the U.S. National Science Foundation under Grant No. OSI-2328774.

DATA AND CODE AVAILABILITY.

The numerical source, reference data, and Lean 4 formalization are provided in the author’s GitHub repository <https://github.com/IKEDAKAZUKI/Quantum-Turing-Pattern>.

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